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Electromagnetic Induction

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 112 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A 30 cm long solenoid has 10 turns per cm and area of $5\ \text{cm}^2$. The current through the solenoid coil varies from 2 A to 4 A in 3.14 s . The e.m.f. induced in the coil is $\alpha \times 10^{-5}$ V. The value α is ____ .

(A) 60 (B) 12 (C) 120 (D) 34

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

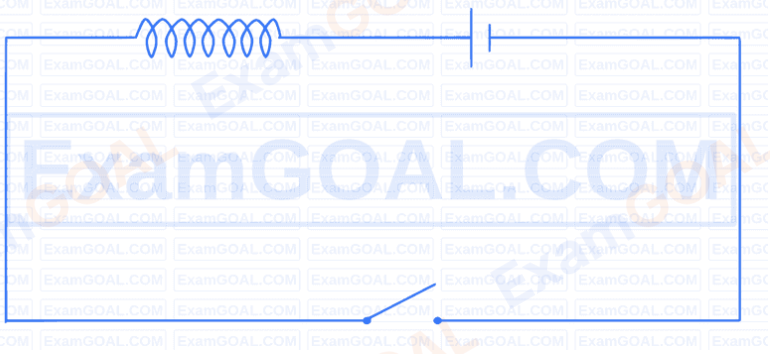
A square loop of side 2 cm is placed in a time varying magnetic field with magnitude as $B = 0.4\sin(300t)$ Tesla. The normal to the plane of loop makes an angle of 60° with the field. The maximum induced emf produced in the loop is ____ mV .

(A) 12 (B) 18 (C) 21 (D) 24

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

An inductor of inductance 10 mH having resistance of 100Ω is connected to battery of E.M.F. 1.0 V through a switch as shown in the figure below. After switch is closed, the ratio of instantaneous voltages across the inductor when the current passing through it is 2 mA and 4 mA is ____ .



(A) 4/3 (B) 3/4 (C) 5/3 (D) 3/5

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

A metal rod of length L rotates about one end at origin with a uniform angular velocity ω. The magnetic field radially falls off as $B(\mathrm{r}) = B_o e^{-\lambda r}$; λ being a positive constant. The emf induced (neglecting the centripetal force on electrons in the rod) is :

(A) $B_o\omega\left[\frac{1}{\lambda^2} - e^{-\lambda L}\left(\frac{1}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

(B) $B_o\omega\left[\frac{1}{\lambda^2} + e^{-\lambda L}\left(\frac{1}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

(C) $B_o\omega\left[\frac{4}{\lambda^2} - e^{-2\lambda L}\left(\frac{1}{\lambda^2} + \frac{2L}{\lambda}\right)\right]$

(D) $B_0\omega\left[\frac{3}{\lambda^2} - e^{-3\lambda L}\left(\frac{3}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

Open solution

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Q5. JEE Main 2026 (Online) 2nd April Evening Shift

A circular current loop of radius R is placed inside square loop of side length L $(L \gg R)$ such that they are co-planar and their centers coincide. The permeability of free space is μ_0. The mutual inductance between circular loop and square loop is _____.

Open solution

(A) $2\sqrt{2}\frac{\mu_0 L^2}{R}$ (B) $\sqrt{2}\frac{\mu_0 L^2}{R}$ (C) $\sqrt{2}\frac{\mu_0 R^2}{L}$ (D) $2\sqrt{2}\frac{\mu_0 R^2}{L}$

Q6. JEE Main 2026 (Online) 2nd April Morning Shift

When a coil is placed in a time dependent magnetic field the power dissipated in it is P. The number of turns, area of the coil and radius of the coil wire are N, A and r respectively. For a second coil number of turns, area of the coil and radius of the coil wire are $2N$, $2A$ and $3r$ respectively. When the first coil is replaced with second coil the power dissipated in it is $\sqrt{2}\alpha P$. The value of α is _____.

Open solution

(A) 36 (B) 128 $\sqrt{2}$ (C) 16 (D) 64

Q7. JEE Main 2026 (Online) 23rd January Evening Shift

A circular loop of radius 7 cm is placed in uniform magnetic field of 0.2 T directed perpendicular to plane of loop. The loop is converted into a square loop in 0.5 s . The EMF induced in the loop is ____ mV.

Open solution

(A) 13.2 (B) 6.6 (C) 1.32 (D) 8.25

Q8. JEE Main 2026 (Online) 23rd January Evening Shift

Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current $I = 10\sin(\omega t)\text{A}$, where $\omega = 1000\text{rad.}/\text{s}$. A circular conducting loop (B) of radius 1 cm coaxially slided through the solenoid at a speed $v = 1\ \text{cm}/\text{s}$. The r.m.s. current through the loop when the coil B is inserted 10 cm inside the solenoid is $\frac{\alpha}{\sqrt{2}}\mu A$. The value of α is ____ .

Open solution

[Resistance of the loop $= 10\Omega$]

(A) 80 (B) 280 (C) 100 (D) 197

Q9. JEE Main 2026 (Online) 23rd January Morning Shift

A 20 m long uniform copper wire held horizontally is allowed to fall under the gravity $\left(g = 10\ \text{m}/\text{s}^2\right)$ through a uniform horizontal magnetic field of 0.5 Gauss perpendicular to the length of the wire. The induced EMF across the wire when it travels a vertical distance of 200 m is ____ mV .

Open solution

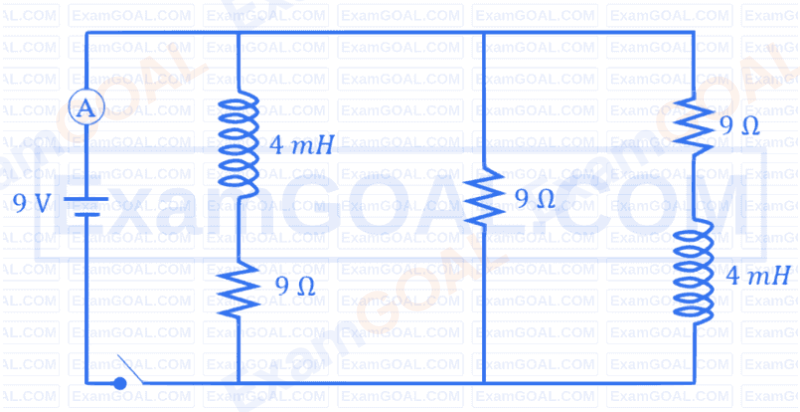
(A) $200\sqrt{10}$ (B) $0.2\sqrt{10}$ (C) $20\sqrt{10}$ (D) $2\sqrt{10}$

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Q10. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Figure shows the circuit that contains three resistances (9Ω each) and two inductors (4 mH each). The reading of ammeter at the moment switch K is turned ON , is ____ A.

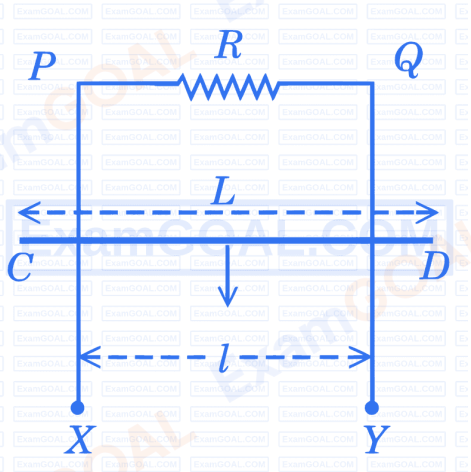


(A) 2 (B) 3 (C) 1 (D) zero

Q11. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

$XPQY$ is a vertical smooth long loop having a total resistance R where PX is parallel to QY and separation between them is l. A constant magnetic field B perpendicular to the plane of the loop exists in the entire space. A rod CD of length $L(L > l)$ and mass m is made to slide down from rest under the gravity as shown in figure. The terminal speed acquired by the rod is ____ $\mathrm{m/s}$. (g = acceleration due to gravity)



(A) $\frac{m\,\mathrm{g}R}{B^2l^2}$ (B) $\frac{2mgR}{B^2L^2}$ (C) $\frac{2mgR}{B^2l^2}$ (D) $\frac{8mgR}{B^2l^2}$

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Q12. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Three identical coils C_1, C_2 and C_3 are closely placed such that they share a common axis. C_2 is exactly midway. C_1 carries current I in anti-clockwise direction while C_3 carries current I in clockwise direction. An induced current flows through C_2 will be in clockwise direction when

(A) C_1 and C_3 move with equal speeds away from C_2

(B) C_1 moves towards C_2 and C_3 moves away from C_2

(C) C_1 and C_3 move with equal speeds towards C_2

(D) C_1 moves away from C_2 and C_3 moves towards C_2

Q13. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

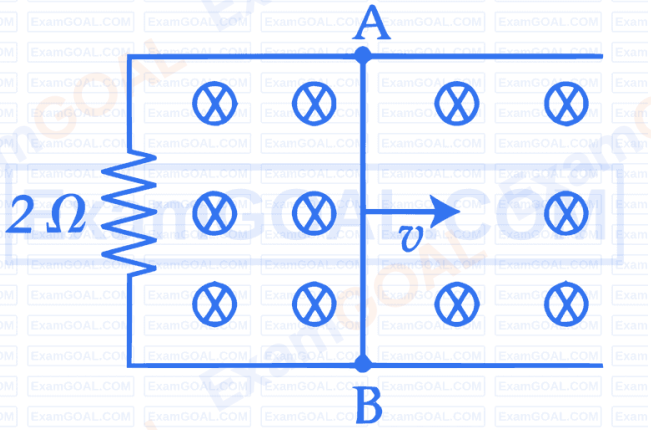
A conducting circular loop of area $1.0\ \mathrm{m}^2$ is placed perpendicular to a magnetic field which varies as $B = \sin(100t)$ Tesla. If the resistance of the loop is 100Ω, then the average thermal energy dissipated in the loop in one period is ____ J.

(A) π^2 (B) $\frac{\pi}{2}$ (C) π (D) 2π

Q14. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T . If the resistance of the total circuit is 2Ω then the force needed to move the rod towards right with constant speed (v) of $1.5\ \mathrm{m/s}$ is ____ N.



(A) 5.7×10^{-2} (B) 7.5×10^{-3} (C) 5.7×10^{-3} (D) 7.5×10^{-2}

Q15. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

A solenoid having area A and length ' l ' is filled with a material having relative permeability 2. The magnetic energy stored in the solenoid is :

(A) $\frac{\mathrm{B}^2\,\mathrm{A}l}{4\mu_0}$ (B) $\frac{\mathrm{B}^2\,\mathrm{A}l}{2\mu_0}$

(C) $\frac{\mathrm{B}^2\,\mathrm{A}l}{\mu_0}$ (D) $\mathrm{B}^2\mathrm{A}l$

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Q16. JEE Main 2025 (Online) 29th January Morning Shift

A coil of area A and N turns is rotating with angular velocity ω in a uniform magnetic field $\vec{B}$ about an axis perpendicular to $\vec{B}$. Magnetic flux φ and induced emf ε across it, at an instant when $\vec{B}$ is parallel to the plane of coil, are :

Open solution

(A) $\varphi = AB, \varepsilon = NAB\omega$ (B) $\varphi = AB, \varepsilon = 0$ (C) $\varphi = 0, \varepsilon = 0$ (D) $\varphi = 0, \varepsilon = NAB\omega$

Q17. JEE Main 2025 (Online) 29th January Morning Shift

Consider I_1 and I_2 are the currents flowing simultaneously in two nearby coils 1 & 2, respectively. If L_1 = self inductance of coil 1, M_{12} = mutual inductance of coil 1 with respect to coil 2, then the value of induced emf in coil 1 will be :

Open solution

(A) $e_1 = -L_1\frac{dI_2}{dt} - M_{12}\frac{dI_1}{dt}$ (B) $e_1 = -L_1\frac{dI_1}{dt} + M_{12}\frac{dI_2}{dt}$

(C) $e_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_1}{dt}$ (D) $e_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_2}{dt}$

Q18. JEE Main 2025 (Online) 28th January Evening Shift

A uniform magnetic field of 0.4 T acts perpendicular to a circular copper disc 20 cm in radius. The disc is having a uniform angular velocity of 10 π rad s^{-1} about an axis through its centre and perpendicular to the disc. What is the potential difference developed between the axis of the disc and the rim? $(\pi = 3.14)$

Open solution

(A) 0.5024 V (B) 0.0628 V (C) 0.2512 V (D) 0.1256 V

Q19. JEE Main 2025 (Online) 23rd January Morning Shift

Regarding self-inductance:

A. The self-inductance of the coil depends on its geometry.

B. Self-inductance does not depend on the permeability of the medium.

C. Self-induced e.m.f. opposes any change in the current in a circuit.

D. Self-inductance is electromagnetic analogue of mass in mechanics.

E. Work needs to be done against self-induced e.m.f. in establishing the current.

Choose the correct answer from the options given below:

Open solution

(A) A, B, C, E only (B) A, B, C, D only (C) A, C, D, E only (D) B, C, D, E only

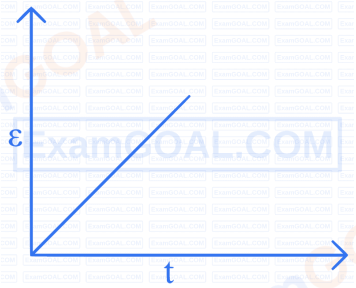
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Q20. JEE Main 2025 (Online) 22nd January Evening Shift

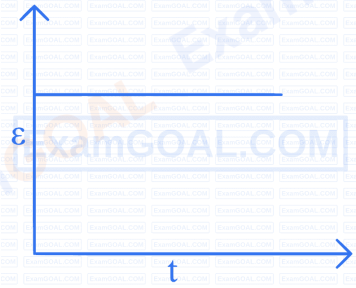
Open solution

A rectangular metallic loop is moving out of a uniform magnetic field region to a field free region with a constant speed. When the loop is partially inside the magnetic field, the plot of magnitude of induced emf (ε) with time (t) is given by

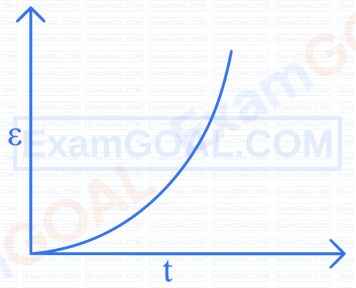
(A)



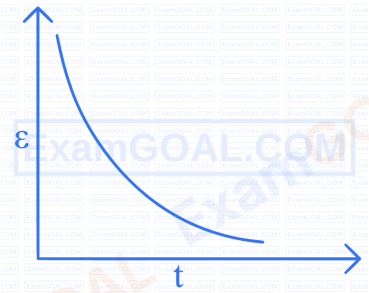
(B)



(C)



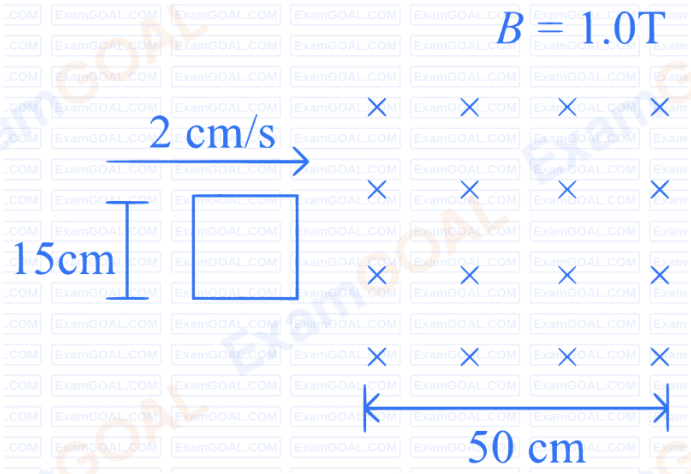
(D)



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Q21. JEE Main 2024 (Online) 9th April Evening Shift

A square loop of side $15\ \text{cm}$ is being moved towards right at a constant speed of $2\ \text{cm/s}$ as shown in figure. The front edge enters the $50\ \text{cm}$ wide magnetic field at $t = 0$. The value of induced emf in the loop at $t = 10\ \text{s}$ will be :



Open solution

(A) zero (B) 4.5 mV (C) 0.3 mV (D) 3 mV

Q22. JEE Main 2024 (Online) 6th April Evening Shift

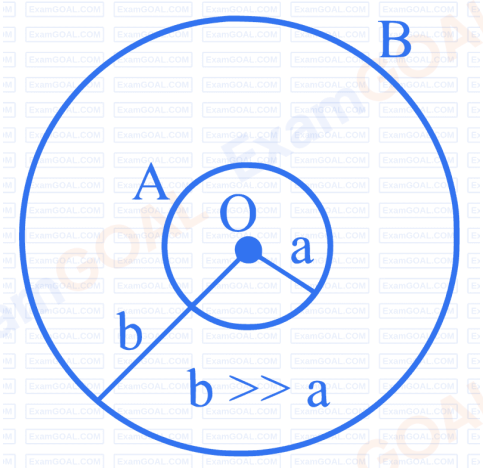
In a coil, the current changes from $-2\ \text{A}$ to $+2\ \text{A}$ in $0.2\ \text{s}$ and induces an emf of $0.1\ \text{V}$. The self inductance of the coil is :

(A) 4 mH (B) 2.5 mH (C) 1 mH (D) 5 mH

Open solution

Q23. JEE Main 2024 (Online) 5th April Morning Shift

Two conducting circular loops A and B are placed in the same plane with their centres coinciding as shown in figure. The mutual inductance between them is:



(A) $\frac{\mu_o}{2\pi} \cdot \frac{b^2}{a}$ (B) $\frac{\mu_o \pi a^2}{2\,b}$

(C) $\frac{\mu_0 \pi b^2}{2a}$ (D) $\frac{\mu_0}{2\pi} \cdot \frac{a^2}{b}$

Open solution

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Q24. JEE Main 2024 (Online) 31st January Morning Shift

A coil is placed perpendicular to a magnetic field of $5000\ \mathrm{T}$. When the field is changed to $3000\ \mathrm{T}$ in $2\ \mathrm{s}$, an induced emf of $22\ \mathrm{V}$ is produced in the coil. If the diameter of the coil is $0.02\ \mathrm{m}$, then the number of turns in the coil is:

Open solution

(A) 35 (B) 70 (C) 7 (D) 140

Q25. JEE Main 2024 (Online) 30th January Evening Shift

Match List I with List II

	List I		List II
(A)	Gauss's law of magnetostatics	(I)	$\oint \vec{E} \cdot \vec{da} = \frac{1}{\varepsilon_0} \int \rho dV$
(B)	Faraday's law of electromagnetic induction	(II)	$\oint \vec{B} \cdot \vec{da} = 0$
(C)	Ampere's law	(III)	$\int \vec{E} \cdot \vec{dl} = \frac{-d}{dt} \int \vec{B} \cdot \vec{da}$
(D)	Gauss's law of electrostatics	(IV)	$\oint \vec{B} \cdot \vec{dl} = \mu_0 I$

Open solution

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II (B) A-IV, B-II, C-III, D-I

(C) A-II, B-III, C-IV, D-I (D) A-I, B-III, C-IV, D-II

Q26. JEE Main 2024 (Online) 29th January Morning Shift

Match List I with List II

	List - I		List - II
(A)	$\oint \vec{B} \cdot \overrightarrow{dl} = \mu_o i_c + \mu_o \varepsilon_o \frac{d\phi_E}{dt}$	(I)	Gauss' law for electricity
(B)	$\oint \vec{E} \cdot \overrightarrow{dl} = \frac{d\phi_B}{dt}$	(II)	Gauss' law for magnetism
(C)	$\oint \vec{E} \cdot \overrightarrow{dA} = \frac{Q}{\varepsilon_o}$	(III)	Faraday law
(D)	$\oint \vec{B} \cdot \overrightarrow{dA} = 0$	(IV)	Ampere - Maxwell law

Open solution

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II (B) A-I, B-II, C-III, D-IV

(C) A-IV, B-I, C-III, D-II (D) A-II, B-III, C-I, D-IV

Q27. JEE Main 2024 (Online) 27th January Morning Shift

A rectangular loop of length $2.5\ \mathrm{m}$ and width $2\ \mathrm{m}$ is placed at 60° to a magnetic field of $4\ \mathrm{T}$. The loop is removed from the field in $10\ \mathrm{sec}$. The average emf induced in the loop during this time is

Open solution

(A) $-2\ \mathrm{V}$ (B) $+2\ \mathrm{V}$ (C) $+1\ \mathrm{V}$ (D) $-1\ \mathrm{V}$

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Q28. JEE Main 2023 (Online) 15th April Morning Shift

$12\ \mathrm{V}$ battery connected to a coil of resistance 6Ω through a switch, drives a constant current in the circuit. The switch is opened in $1\ \mathrm{ms}$. The emf induced across the coil is $20\ \mathrm{V}$. The inductance of the coil is :

Open solution

(A) $5\ \mathrm{mH}$ (B) $8\ \mathrm{mH}$ (C) $10\ \mathrm{mH}$ (D) $12\ \mathrm{mH}$

Q29. JEE Main 2023 (Online) 11th April Evening Shift

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Open solution

Assertion A: A bar magnet dropped through a metallic cylindrical pipe takes more time to come down compared to a non-magnetic bar with same geometry and mass.

Reason R: For the magnetic bar, Eddy currents are produced in the metallic pipe which oppose the motion of the magnetic bar.

In the light of the above statements, choose the correct answer from the options given below

(A) A is true but R is false

(B) A is false but R is true

(C) Both A and R are true but R is NOT the correct explanation of A

(D) Both A and R are true and R is the correct explanation of A

Q30. JEE Main 2023 (Online) 10th April Evening Shift

A bar magnet is released from rest along the axis of a very long vertical copper tube. After some time the magnet will

Open solution

(A) move down with almost constant speed

(B) move down with an acceleration equal to g

(C) move down with an acceleration greater than g

(D) oscillate inside the tube

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Q31. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

Given below are two statements:

Statement I : If the number of turns in the coil of a moving coil galvanometer is doubled then the current sensitivity becomes double.

Statement II : Increasing current sensitivity of a moving coil galvanometer by only increasing the number of turns in the coil will also increase its voltage sensitivity in the same ratio

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q32. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

An emf of $0.08\ \text{V}$ is induced in a metal rod of length $10\ \text{cm}$ held normal to a uniform magnetic field of $0.4\ \text{T}$, when moves with a velocity of:

(A) $20\ \text{ms}^{-1}$ (B) $2\ \text{ms}^{-1}$ (C) $3.2\ \text{ms}^{-1}$ (D) $0.5\ \text{ms}^{-1}$

Q33. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

Certain galvanometers have a fixed core made of non magnetic metallic material. The function of this metallic material is

(A) to bring the coil to rest quickly

(B) to produce large deflecting torque on the coil

(C) to oscillate the coil in magnetic field for longer period of time

(D) to make the magnetic field radial

Q34. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

The induced emf can be produced in a coil by

A. moving the coil with uniform speed inside uniform magnetic field

B. moving the coil with non uniform speed inside uniform magnetic field

C. rotating the coil inside the uniform magnetic field

D. changing the area of the coil inside the uniform magnetic field

Choose the correct answer from the options given below:

(A) A and C only (B) C and D only (C) B and D only (D) B and C only

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Q35. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

A coil is placed in magnetic field such that plane of coil is perpendicular to the direction of magnetic field. The magnetic flux through a coil can be changed :

A. By changing the magnitude of the magnetic field within the coil.

B. By changing the area of coil within the magnetic field.

C. By changing the angle between the direction of magnetic field and the plane of the coil.

D. By reversing the magnetic field direction abruptly without changing its magnitude.

Choose the most appropriate answer from the options given below :

(A) A, B and D only (B) A, B and C only (C) A and B only (D) A and C only

Q36. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

Spherical insulating ball and a spherical metallic ball of same size and mass are dropped from the same height. Choose the correct statement out of the following

{Assume negligible air friction}

(A) Metal ball will reach the earth's surface earlier than the insulating ball

(B) Both will reach the earth's surface simultaneously.

(C) Insulating ball will reach the earth's surface earlier than the metal ball

(D) Time taken by them to reach the earth's surface will be independent of the properties of their materials

Q37. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

A square loop of area 25 cm^2 has a resistance of 10 Ω. The loop is placed in uniform magnetic field of magnitude 40.0 T. The plane of loop is perpendicular to the magnetic field. The work done in pulling the loop out of the magnetic field slowly and uniformly in 1.0 sec, will be

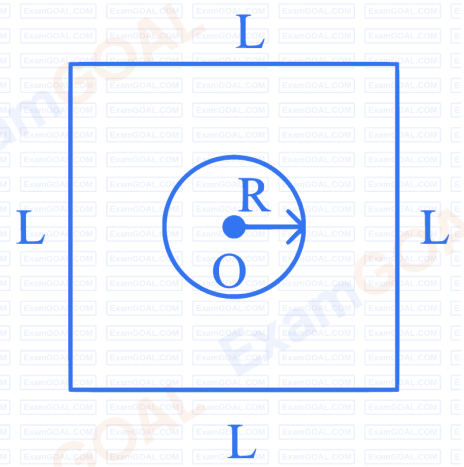
(A) 1.0×10^{-3} J (B) 5×10^{-3} J (C) 2.5×10^{-3} J (D) 1.0×10^{-4} J

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Q38. JEE Main 2023 (Online) 29th January Morning Shift

Find the mutual inductance in the arrangement, when a small circular loop of wire of radius 'R' is placed inside a large square loop of wire of side L ($L \gg R$). The loops are coplanar and their centres coincide :

Open solution



(A) $M = \frac{\sqrt{2}\mu_0 R}{L^2}$

(B) $M = \frac{2\sqrt{2}\mu_0 R}{L^2}$

(C) $M = \frac{2\sqrt{2}\mu_0 R^2}{L}$

(D) $M = \frac{\sqrt{2}\mu_0 R^2}{L}$

Q39. JEE Main 2023 (Online) 25th January Evening Shift

A wire of length 1m moving with velocity 8 m/s at right angles to a magnetic field of 2T. The magnitude of induced emf, between the ends of wire will be _____.

Open solution

(A) 20 V

(B) 8 V

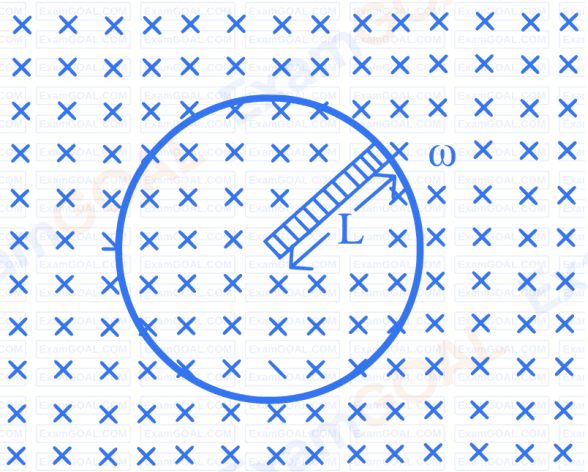
(C) 16 V

(D) 12 V

Q40. JEE Main 2023 (Online) 24th January Evening Shift

A metallic rod of length 'L' is rotated with an angular speed of 'ω' normal to a uniform magnetic field 'B' about an axis passing through one end of rod as shown in figure. The induced emf will be :

Open solution



(A) $\frac{1}{2}B^2L^2\omega$

(B) $\frac{1}{2}BL^2\omega$

(C) $\frac{1}{4}BL^2\omega$

(D) $\frac{1}{4}B^2L\omega$

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Q41. JEE Main 2023 (Online) 24th January Morning Shift

A conducting circular loop of radius $\frac{10}{\sqrt{\pi}}$ cm is placed perpendicular to a uniform magnetic field of 0.5 T. The magnetic field is decreased to zero in 0.5 s at a steady rate. The induced emf in the circular loop at 0.25 s is :

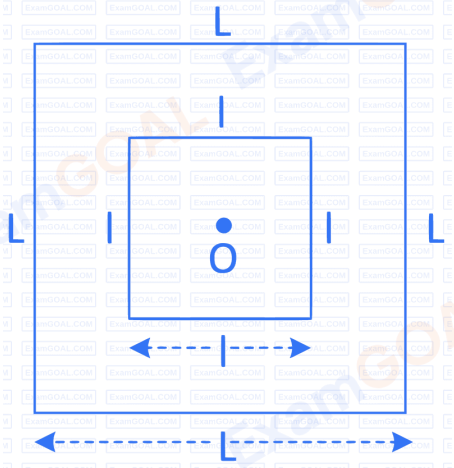
Open solution

(A) emf = 10 mV (B) emf = 5 mV (C) emf = 100 mV (D) emf = 1 mV

Q42. JEE Main 2022 (Online) 25th July Morning Shift

A small square loop of wire of side l is placed inside a large square loop of wire $\mathrm{L}(\mathrm{L} >> l)$. Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is :

Open solution



(A) $\frac{2\sqrt{2}\mu_0 \mathrm{L}^2}{\pi l}$ (B) $\frac{\mu_0 l^2}{2\sqrt{2}\pi \mathrm{L}}$

(C) $\frac{2\sqrt{2}\mu_0 l^2}{\pi \mathrm{L}}$ (D) $\frac{\mu_0 \mathrm{L}^2}{2\sqrt{2}\pi l}$

Q43. JEE Main 2022 (Online) 28th June Evening Shift

A coil is placed in a time varying magnetic field. If the number of turns in the coil were to be halved and the radius of wire doubled, the electrical power dissipated due to the current induced in the coil would be :

Open solution

(Assume the coil to be short circuited.)

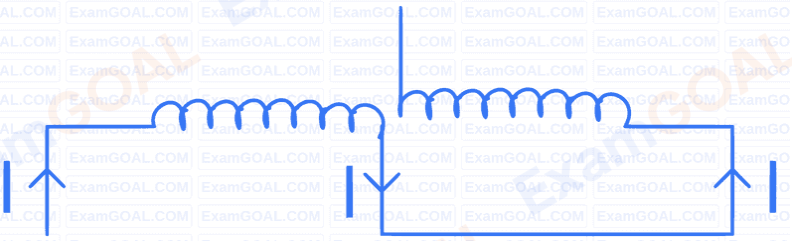
(A) Halved (B) Quadrupled (C) The same (D) Doubled

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Q44. JEE Main 2022 (Online) 26th June Evening Shift

Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M. The equivalent self inductance of the combination will be :

Open solution



(A) $\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{M}$

(B) L_1+L_2+M

(C) L_1+L_2+2M

(D) L_1+L_2-2M

Q45. JEE Main 2022 (Online) 26th June Evening Shift

A metallic conductor of length 1 m rotates in a vertical plane parallel to east-west direction about one of its end with angular velocity 5 rad s^{-1}. If the horizontal component of earth's magnetic field is 0.2×10^{-4} T, then emf induced between the two ends of the conductor is :

Open solution

(A) 5 μV

(B) 50 μV

(C) 5 mV

(D) 50 mv

Q46. JEE Main 2022 (Online) 26th June Morning Shift

The magnetic flux through a coil perpendicular to its plane is varying according to the relation $\phi = (5t^3 + 4t^2 + 2t - 5)$ Weber. If the resistance of the coil is 5 ohm, then the induced current through the coil at t = 2 s will be,

Open solution

(A) 15.6 A

(B) 16.6 A

(C) 17.6 A

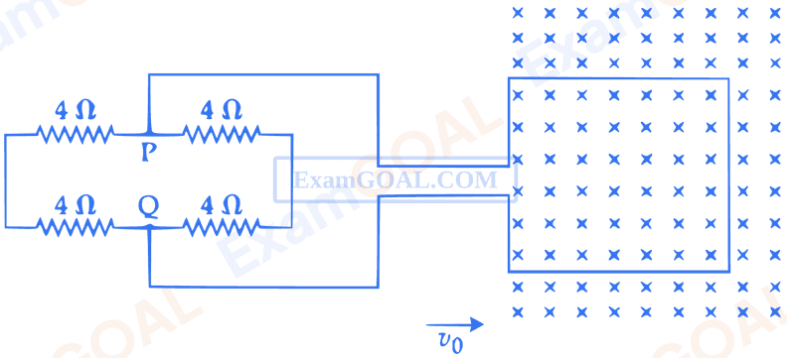
(D) 18.6 A

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Q47. JEE Main 2021 (Online) 1st September Evening Shift

Open solution

A square loop of side 20 cm and resistance 1Ω is moved towards right with a constant speed v_0. The right arm of the loop is in a uniform magnetic field of 5T. The field is perpendicular to the plane of the loop and is going into it. The loop is connected to a network of resistors each of value 4Ω. What should be the value of v_0 so that a steady current of 2 mA flows in the loop?

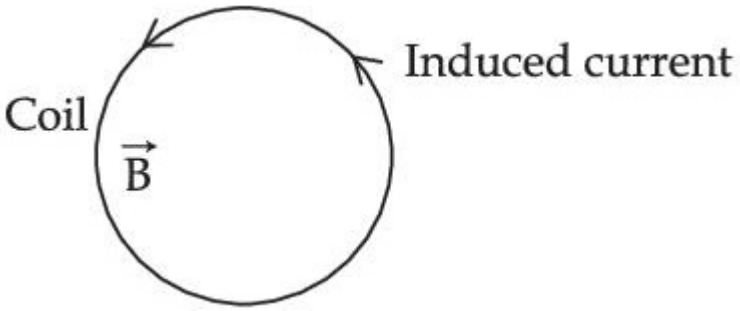


(A) 1 m/s (B) 1 cm/s (C) 10^2 m/s (D) 10^{-2} cm/s

Q48. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

A coil is placed in a magnetic field $\vec{B}$ as shown below :



A current is induced in the coil because $\vec{B}$ is :

(A) Outward and decreasing with time
(B) Parallel to the plane of coil and decreasing with time
(C) Outward and increasing with time
(D) Parallel to the plane of coil and increasing with time

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Q49. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

A small square loop of side 'a' and one turn is placed inside a larger square loop of side b and one turn (b >> a). The two loops are coplanar with their centres coinciding. If a current I is passed in the square loop of side 'b', then the coefficient of mutual inductance between the two loops is :

(A) $\frac{\mu_0}{4\pi} 8\sqrt{2} \frac{a^2}{b}$

(B) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{a}$

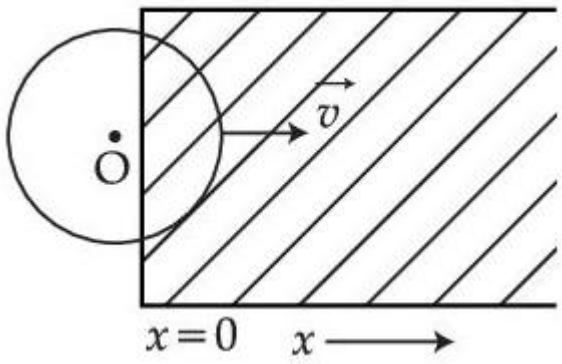
(C) $\frac{\mu_0}{4\pi} 8\sqrt{2} \frac{b^2}{a}$

(D) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{b}$

Q50. JEE Main 2021 (Online) 27th August Evening Shift

Open solution

A constant magnetic field of 1T is applied in the x > 0 region. A metallic circular ring of radius 1m is moving with a constant velocity of 1 m/s along the x-axis. At t = 0s, the centre of O of the ring is at x = −1m. What will be the value of the induced emf in the ring at t = 1s? (Assume the velocity of the ring does not change.)



(A) 1V

(B) 2πV

(C) 2V

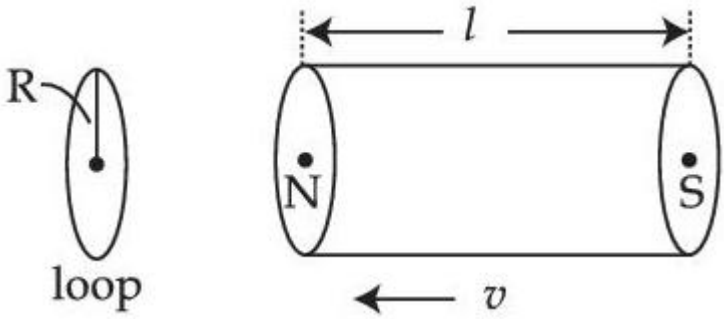
(D) 0V

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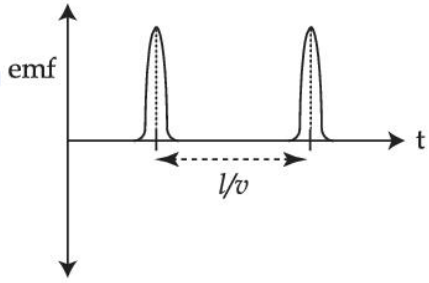
Q51. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

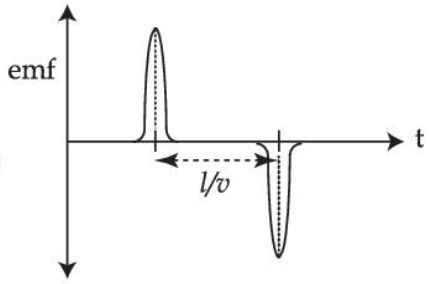
A bar magnet is passing through a conducting loop of radius R with velocity v. The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve :



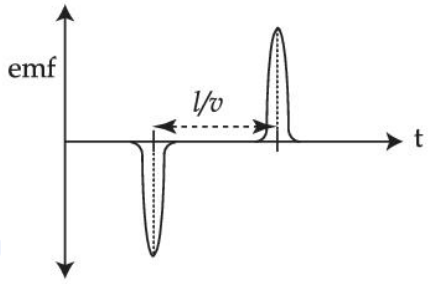
(A)



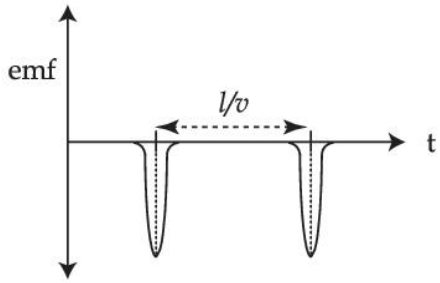
(B)



(C)



(D)



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Q52. JEE Main 2021 (Online) 26th August Morning Shift

An inductor coil stores 64 J of magnetic field energy and dissipates energy at the rate of 640 W when a current of 8A is passed through it. If this coil is joined across an ideal battery, find the time constant of the circuit in seconds :

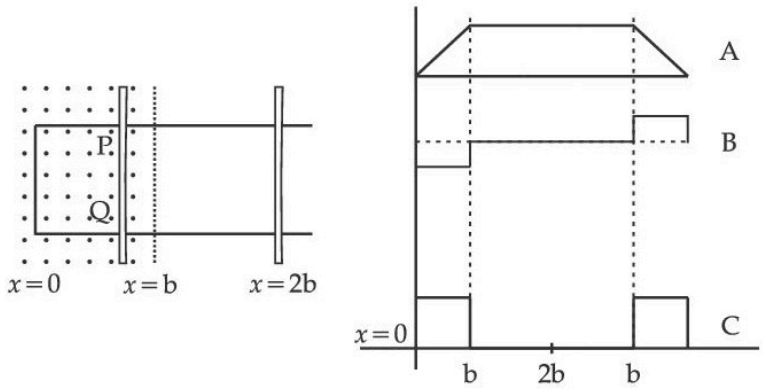
Open solution

(A) 0.4 (B) 0.8 (C) 0.125 (D) 0.2

Q53. JEE Main 2021 (Online) 20th July Morning Shift

The arm PQ of a rectangular conductor is moving from x = 0 to x = 2b outwards and then inwards from x = 2b to x = 0 as shown in the figure. A uniform magnetic field perpendicular to the plane is acting from x = 0 to x = b. Identify the graph showing the variation of different quantities with distance.

Open solution



(A) A-Flux, B-Power dissipated, C-EMF (B) A-Flux, B-EMF, C-Power dissipated

(C) A-Power dissipated, B-Flux, C-EMF (D) A-EMF, B-Power dissipated, C-Flux

Q54. JEE Main 2021 (Online) 18th March Evening Shift

The time taken for the magnetic energy to reach 25% of its maximum value, when a solenoid of resistance R, inductance L is connected to a battery, is :

Open solution

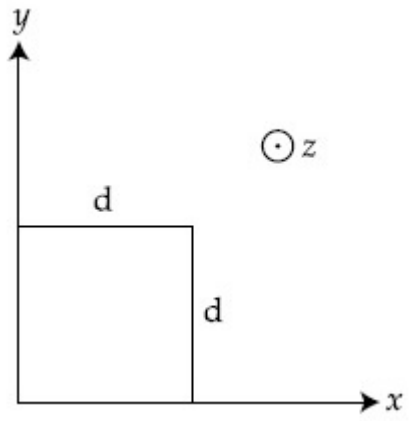
(A) infinite (B) $\frac{L}{R}$ ln10 (C) $\frac{L}{R}$ ln2 (D) $\frac{L}{R}$ ln5

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Q55. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

The magnetic field in a region is given by $\vec{B} = B_o\left(\frac{x}{a}\right)\hat{k}$. A square loop of side d is placed with its edges along the x and y axes. The loop is moved with a constant velocity $\vec{v} = v_0\hat{i}$. The emf induced in the loop is :

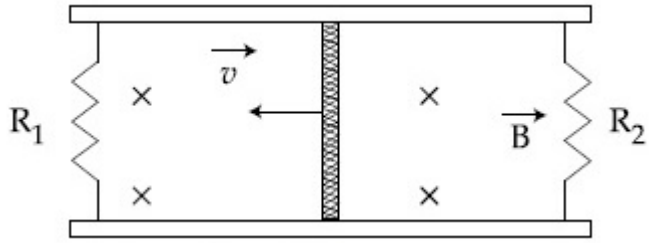


(A) $\frac{B_o v_o d^2}{2a}$ (B) $\frac{B_o v_o^2 d}{2a}$ (C) $\frac{B_o v_o d}{2a}$ (D) $\frac{B_o v_o d^2}{a}$

Q56. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

A conducting bar of length L is free to slide on two parallel conducting rails as shown in the figure



Two resistors R_1 and R_2 are connected across the ends of the rails. There is a uniform magnetic field $\vec{B}$ pointing into the page. An external agent pulls the bar to the left at a constant speed v.

The correct statement about the directions of induced currents I_1 and I_2 flowing through R_1 and R_2 respectively is :

(A) Both I_1 and I_2 are in clockwise direction

(B) I_1 is in clockwise direction and I_2 is in anticlockwise direction

(C) I_1 is in anticlockwise direction and I_2 is in clockwise direction

(D) Both I_1 and I_2 are in anticlockwise direction

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Q57. JEE Main 2021 (Online) 26th February Evening Shift

An aeroplane, with its wings spread 10 m, is flying at a speed of 180 km/h in a horizontal direction. The total intensity of earth's field at that part is 2.5×10^{-4} Wb/m^2 and the angle of dip is 60°. The emf induced between the tips of the plane wings will be _____.

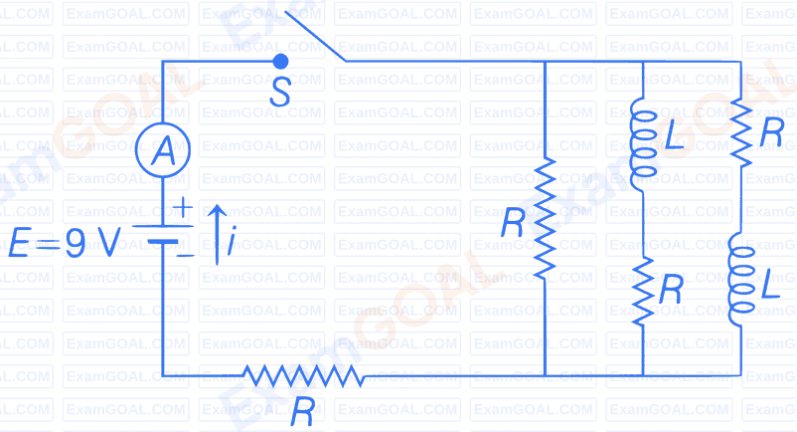
Open solution

(A) 88.37 mV (B) 62.50 mV (C) 54.125 mV (D) 108.25 mV

Q58. JEE Main 2021 (Online) 24th February Evening Shift

Figure shows a circuit that contains four identical resistors with resistance R = 2.0Ω, two identical inductors with inductance L = 2.0 mH and an ideal battery with emf E = 9V. The current 'i' just after the switch 'S' is closed will be :

Open solution



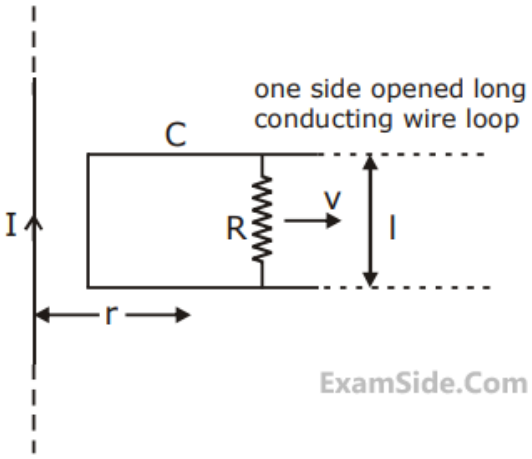
(A) 3.0 A (B) 3.37 A (C) 9 A (D) 2.25 A

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Q59. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

An infinitely long, straight wire carrying current I, one side opened rectangular loop and a conductor C with a sliding connector are located in the same plane, as shown, in the figure. The connector has length l and resistance R. It slides to the right with a velocity v. The resistance of the conductor and the self inductance of the loop are negligible. The induced current in the loop, as a function of separation r, between the connector and the straight wire is :



(A) $\frac{\mu_0}{4\pi}\frac{Ivl}{Rr}$

(B) $\frac{\mu_0}{\pi}\frac{Ivl}{Rr}$

(C) $\frac{\mu_0}{2\pi}\frac{Ivl}{Rr}$

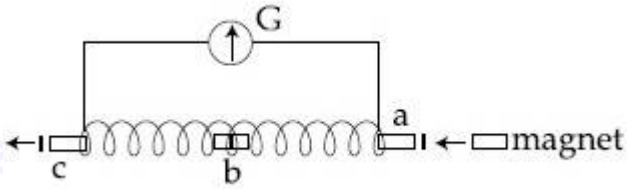
(D) $\frac{2\mu_0}{\pi}\frac{Ivl}{Rr}$

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Q60. JEE Main 2020 (Online) 4th September Morning Slot

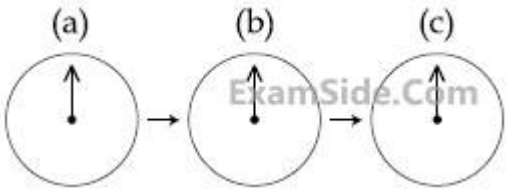
A small bar magnet is moved through a coil at constant speed from one end to the other. Which of the following series of observations will be seen on the galvanometer G attached across the coil?

Open solution

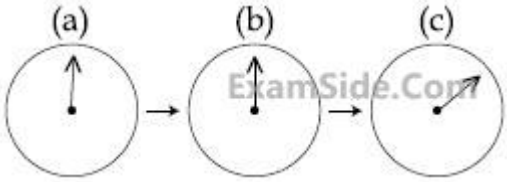


Three positions shown describe :
(a) the magnet's entry
(b) magnet is completely inside and
(c) magnet's exit.

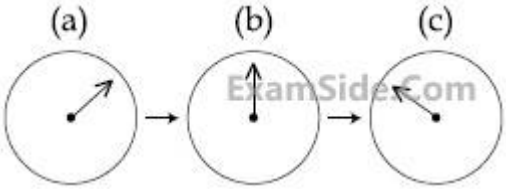
(A)



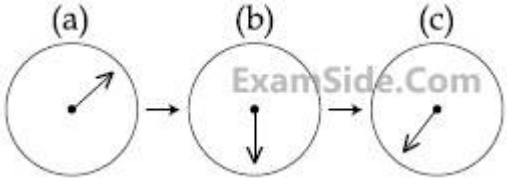
(B)



(C)



(D)



Q61. JEE Main 2020 (Online) 3rd September Evening Slot

A uniform magnetic field B exists in a direction perpendicular to the plane of a square loop made of a metal wire. The wire has a diameter of 4 mm and a total length of 30 cm. The magnetic field changes with time at a steady rate $\frac{dB}{dt} = 0.032\ \text{Ts}^{-1}$. The induced current in the loop is close to (Resistivity of the metal wire is $1.23 \times 10^{-8}\ \Omega\text{m}$)

Open solution

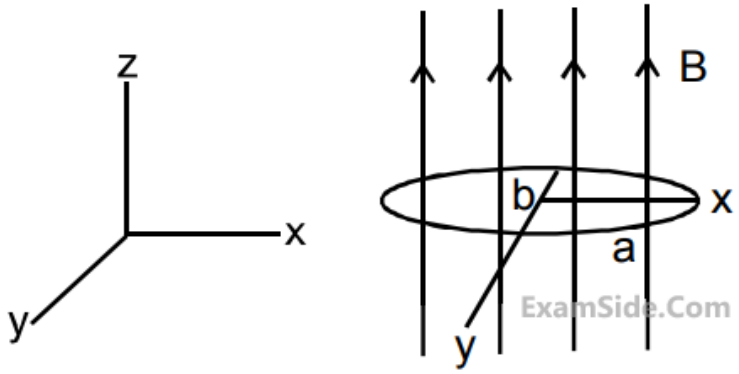
(A) 0.53 A (B) 0.43 A (C) 0.34 A (D) 0.61 A

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Q62. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

An elliptical loop having resistance R, of semi major axis a, and semi minor axis b is placed in magnetic field as shown in the figure. If the loop is rotated about the x-axis with angular frequency ω, the average power loss in the loop due to Joule heating is :



(A) $\frac{\pi abB\omega}{R}$

(B) $\frac{\pi^2a^2b^2B^2\omega^2}{R}$

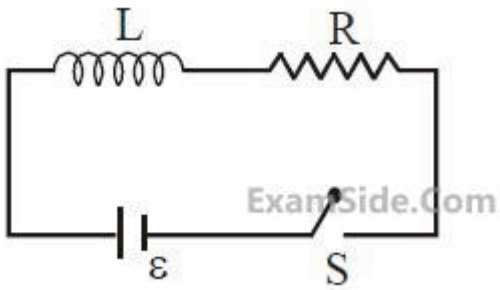
(C) $\frac{\pi^2a^2b^2B^2\omega^2}{2R}$

(D) Zero

Q63. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

A shown in the figure, a battery of emf ε is connected to an inductor L and resistance R in series. The switch is closed at t = 0. The total charge that flows from the battery, between t = 0 and t = t_c (t_c is the time constant of the circuit) is :



(A) $\frac{\varepsilon L}{eR^2}$

(B) $\frac{\varepsilon L}{R^2}$

(C) $\frac{\varepsilon L}{R^2}\left(1-\frac{1}{e}\right)$

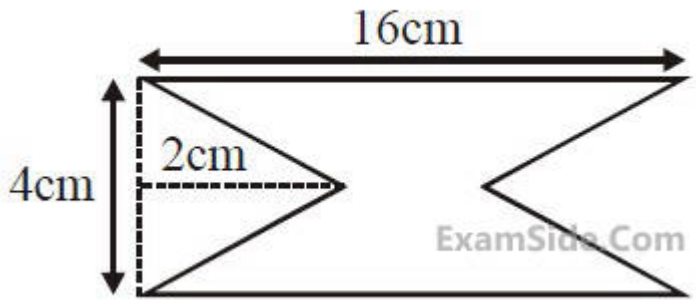
(D) $\frac{\varepsilon R}{eL^2}$

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Q64. JEE Main 2020 (Online) 8th January Morning Slot

At time t = 0 magnetic field of 1000 Gauss is passing perpendicularly through the area defined by the closed loop shown in the figure. If the magnetic field reduces linearly to 500 Gauss, in the next 5s, then induced EMF in the loop is :

Open solution



(A) 48 μV (B) 28 μV (C) 56 μV (D) 36 μV

Q65. JEE Main 2020 (Online) 7th January Evening Slot

A planar loop of wire rotates in a uniform magnetic field. Initially at t = 0, the plane of the loop is perpendicular to the magnetic field. If it rotates with a period of 10 s about an axis in its plane then the magnitude of induced emf will be maximum and minimum, respectively at :

Open solution

(A) 2.5 s and 7.5 s (B) 5.0 s and 10.0 s (C) 5.0 s and 7.5 s (D) 2.5 s and 5.0 s

Q66. JEE Main 2020 (Online) 7th January Morning Slot

Consider a circular coil of wire carrying constant current I, forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by ϕ_i. The magnetic flux through the area of the circular coil area is given by ϕ_0. Which of the following option is correct?

Open solution

(A) $\phi_i = \phi_0$ (B) $\phi_i < \phi_0$ (C) $\phi_i > \phi_0$ (D) $\phi_i = -\phi_0$

Q67. JEE Main 2020 (Online) 7th January Morning Slot

A long solenoid of radius R carries a time (t) - dependent current $I(t)=I_0t(1 - t)$. A ring of radius 2R is placed coaxially near its middle. During the time interval $0 \leq t \leq 1$, the induced current (I_R) and the induced EMF(V_R) in the ring change as :

Open solution

(A) Direction of I_R remains unchanged and V_R is zero at t = 0.25

(B) Direction of I_R remains unchanged and V_R is maximum at t = 0.5

(C) At t = 0.25 direction of I_R reverses and V_R is maximum

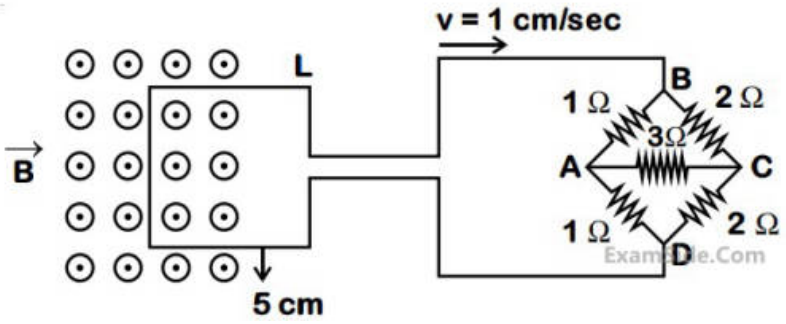
(D) At t = 0.5 direction of I_R reverses and V_R is zero

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Q68. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

The figure shows a square loop L of side 5 cm which is connected to a network of resistances. The whole setup is moving towards right with a constant speed of 1 cm s^{-1}. At some instant, a part of L is in a uniform magnetic field of 1 T, perpendicular to the plane of the loop. If the resistance of L is 1.7 Ω, the current in the loop at that instant will be close to :



(A) 115 μA (B) 170 μA (C) 60 μA (D) 150 μA

Q69. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

Two coils 'P' and 'Q' are separated by some distance. When a current of 3 A flows through coil 'P', a magnetic flux of 10^{-3} Wb passes through 'Q'. No current is passed through 'Q'. When no current passes through 'P' and a current of 2 A passes through 'Q', the flux through 'P' is :-

(A) 3.67×10^{-4} Wb (B) 3.67×10^{-3} Wb (C) 6.67×10^{-4} Wb (D) 6.67×10^{-3} Wb

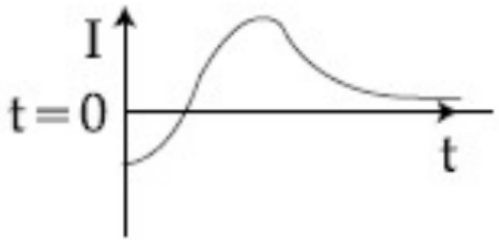
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Q70. JEE Main 2019 (Online) 9th April Evening Slot

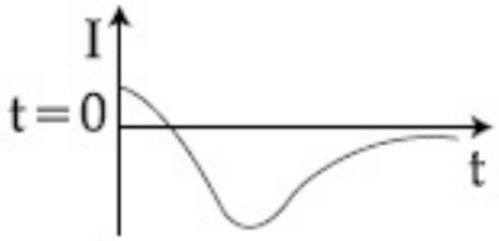
Open solution

A very long solenoid of radius R is carrying current $I(t) = kte^{-at}(k > 0)$, as a function of time ($t \geq 0$). counter clockwise current is taken to be positive. A circular conducting coil of radius 2R is placed in the equatorial plane of the solenoid and concentric with the solenoid. The current induced in the outer coil is correctly depicted, as a function of time, by :-

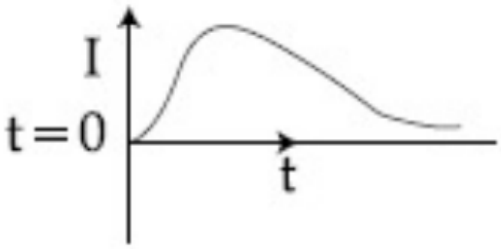
(A)



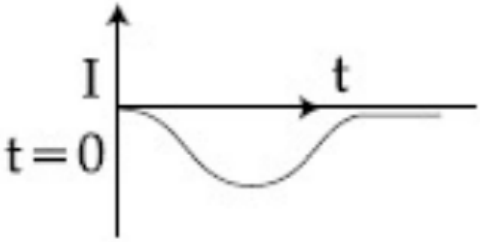
(B)



(C)



(D)



Q71. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

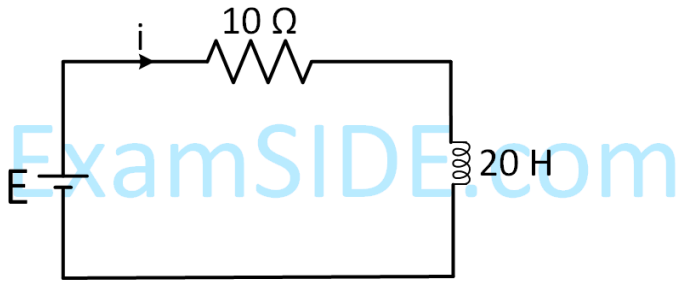
The total number of turns and cross-section area in a solenoid is fixed. However, its length L is varied by adjusting the separation between windings. The inductance of solenoid will be proportional to :

(A) $1/L^2$ (B) $1/L$ (C) L (D) L^2

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Q72. JEE Main 2019 (Online) 8th April Morning Slot

A 20 Henry inductor coil is connected to a 10 ohm resistance in series as shown in figure. The time at which rate of dissipation of energy (joule's heat) across resistance is equal to the rate at which magnetic energy is stored in the inductor is :



Open solution

(A) $\frac{2}{\ln 2}$ (B) $\ln 2$ (C) $2\ln 2$ (D) $\frac{1}{2}\ln 2$

Q73. JEE Main 2019 (Online) 12th January Evening Slot

A 10 m long horizontal wire extends from North East to South West. It is falling with a speed of 5.0 ms^{-1}, at right angles to the horizontal component of the earth's magnetic field of 0.3×10^{-4} Wb/m^2. The value of the induced emf in wire is :

Open solution

(A) 0.3×10^{-3} V (B) 2.5×10^{-3} V (C) 1.5×10^{-3} V (D) 1.1×10^{-3} V

Q74. JEE Main 2019 (Online) 11th January Evening Slot

A copper wire is wound on a wooden frame, whose shape is that of an equilateral triangle. If the linear dimension of each side of the frame is increased by a factor of 3, keeping the number of turns of the coil per unit length of the frame the same, then the self inductance of the coil:

Open solution

(A) decreases by a factor of $9\sqrt{3}$
(B) increases by a factor of 27
(C) decreases by a factor of 9
(D) increases by a factor of 3

Q75. JEE Main 2019 (Online) 11th January Morning Slot

There are two long co-axial solenoids of same length l. The inner and outer coils have radii r_1 and r_2 and number of turns per unit length n_1 and n_2, respectively. The ratio of mutual inductance to the self - inductance of the inner-coil is :

Open solution

(A) $\frac{n_2}{n_1} \cdot \frac{{r_2}^2}{{r_1}^2}$
(B) $\frac{n_2}{n_1}$
(C) $\frac{n_1}{n_2}$
(D) $\frac{n_2}{n_1} \cdot \frac{r_1}{r_2}$

Q76. JEE Main 2019 (Online) 10th January Evening Slot

The self induced emf of a coil is 25 volts. When the current in it is changed at uniform rate from 10 A to 25 A in 1s, the change in the energy of the inductance is -

Open solution

(A) 740 J (B) 637.5 J (C) 540 J (D) 437.5 J

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Q77. JEE Main 2019 (Online) 10th January Morning Slot

A solid metal cube of edge length 2 cm is moving in a positive y-direction at a constant speed of 6 m/s. There is a uniform magnetic field of 0.1 T in the positive z-direction. The potential difference between the two faces of the cube perpendicular to the x-axis, is -

Open solution

(A) 2mV (B) 12 mV (C) 6 mV (D) 1 mV

Q78. JEE Main 2019 (Online) 9th January Morning Slot

A conducting circular loop made of a thin wire, has area 3.5×10^{-3} m^2 and resistance 10 Ω. It is placed perpendicular to a time dependent magnetic field B(t) = (0.4T)sin(50πt). The field is uniform in space. Then the net charge flowing through the loop during t = 0 s and t = 10 ms is close to :

Open solution

(A) 0.14 mC (B) 0.7 mC (C) 0.21 mC (D) 0.6 mC

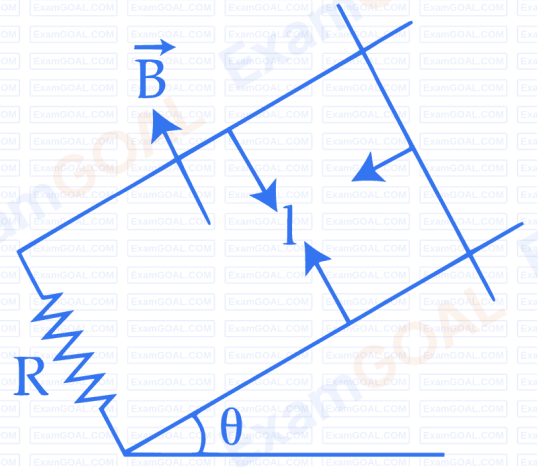
Q79. JEE Main 2018 (Online) 16th April Morning Slot

A coil of cross-sectional area A having n turns is placed in a uniform magnetic field B. When it is rotated with an angular velocity ω, the maxium e.m.f. induced in the coil will be:

Open solution

(A) 3 nBAω (B) $\frac{3}{2}$ nBAω (C) nBAω (D) $\frac{1}{2}$ nBAω

Q80. JEE Main 2018 (Online) 15th April Evening Slot



Open solution

A copper rod of mass m slides under gravity on two smooth parallel rails, with separation *l* and set at an angle of θ with the horizontal. At the bottom rails are joined by a resistance R. There is a uniform magnetic field B normal to the plane of the rails, as shown in the igure. The terminal speed of the copper rod is :

(A) $\frac{mgR\tan\theta}{B^2l^2}$ (B) $\frac{mgR\cot\theta}{B^2l^2}$

(C) $\frac{mgR\sin\theta}{B^2l^2}$ (D) $\frac{mgR\cos\theta}{B^2l^2}$

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Q81. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

At the center of a fixed large circular coil of radius R, a much smaller circular coil of radius r is placed. The two coils are concentric and are in the same plane. The larger coil carries a current I. The smaller coil is set to rotate with a constant angular velocity ω about an axis along their common diameter. Calculate the emf induced in the smaller coil after a time t of its start of rotation.

(A) $\frac{\mu_o I}{2R}\,\omega\,\pi\,r^2\sin\omega\,t$

(B) $\frac{\mu_o I}{4R}\,\omega\,\pi\,r^2\sin\omega\,t$

(C) $\frac{\mu_o I}{4R}\,\omega\,r^2\sin\omega\,t$

(D) $\frac{\mu_o I}{2R}\,\omega\,r^2\sin\omega\,t$

Q82. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

A small circular loop of wire of radius a is located at the centre of a much larger circular wire loop of radius b. The two loops are in the same plane. The outer loop of radius b carries an alternating current $I = I_0 \cos(\omega t)$. The emf induced in the smaller inner loop is nearly :

(A) $\frac{\pi\mu_o I_o}{2}\cdot\frac{a^2}{b}\,\omega\sin(\omega t)$

(B) $\frac{\pi\mu_o I_o}{2}\cdot\frac{a^2}{b}\,\omega\cos(\omega t)$

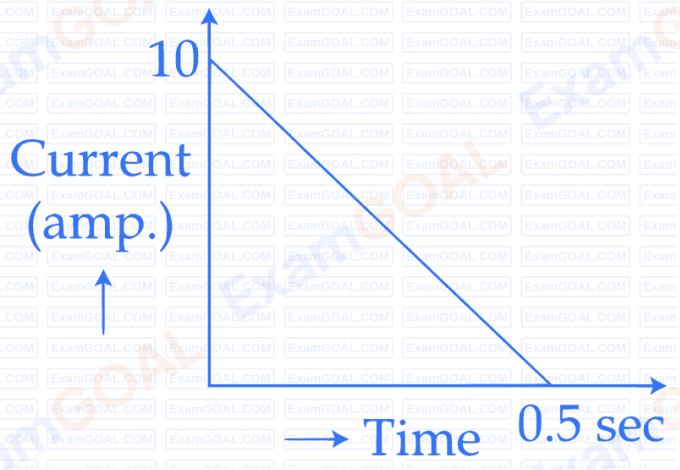
(C) $\pi\mu_o I_o\,\frac{a^2}{b}\omega\sin(\omega t)$

(D) $\frac{\pi\mu_o I_o\,b^2}{a}\omega\cos(\omega t)$

Q83. JEE Main 2017 (Offline)

Open solution

In a coil of resistance 100 Ω, a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is:



(A) 275 Wb

(B) 200 Wb

(C) 225 Wb

(D) 250 Wb

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Q84. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

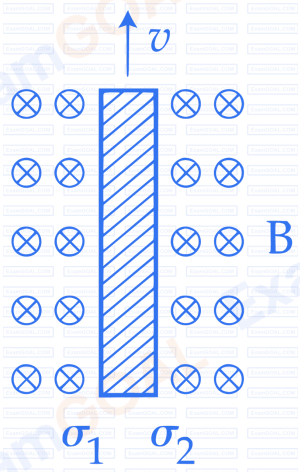
A conducting metal circular-wire-loop of radius r is placed perpendicular to a magnetic field which varies with time as
$B = B_0 e^{\frac{-t}{\tau}}$, where B_0 and τ are constants, at time t = 0. If the resistance of the loop is R then the heat generated in the loop after a long time ($t \to \infty$) is :

(A) $\frac{\pi^2 r^4 B_0^4}{2\tau R}$
(B) $\frac{\pi^2 r^4 B_0^2}{2\tau R}$
(C) $\frac{\pi^2 r^4 B_0^2 R}{\tau}$
(D) $\frac{\pi^2 r^4 B_0^2}{\tau R}$

Q85. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

Consider a thin metallic sheet perpendicular to the plane of the paper moving with speed *'v'* in a uniform magnetic field B going into the plane of the paper (See figure). If charge densities σ_1 and σ_2 are induced on the left and right surfaces, respectively, of the sheet then (ignore fringe effects) :



(A) $\sigma_1 = \in_0 v\, B$, $\sigma_2 = -\in_0 v\, B$
(B) $\sigma_1 = \frac{\in_0 v\, B}{2}$, $\sigma_2 = \frac{-\in_0 vB}{2}$
(C) $\sigma_1 = \sigma_2 = \in_0 vB$
(D) $\sigma_1 = \frac{-\in_0 vB}{2}$, $\sigma_2 = \frac{\in_0 vB}{2}$,

Q86. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

When current in a coil changes from 5 A to 2 A in 0.1 s, an average voltage of 50 V is produced. The self - inductance of the coil is :

(A) 0.67 H
(B) 1.67 H
(C) 3 H
(D) 6 H

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Q87. JEE Main 2015 (Offline)

Open solution

Two coaxial solenoids of different radius carry current I in the same direction. $\overrightarrow{F_1}$ be the magnetic force on the inner solenoid due to the outer one and $\overrightarrow{F_2}$ be the magnetic force on the outer solenoid due to the inner one. Then :

(A) $\overrightarrow{F_1}$ is radially in wards and $\overrightarrow{F_2} = 0$

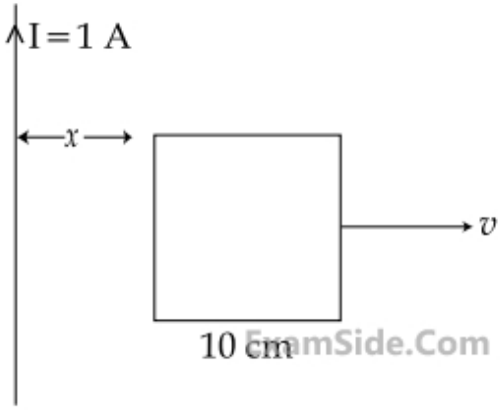
(B) $\overrightarrow{F_1}$ is radially outwards and $\overrightarrow{F_2} = 0$

(C) $\overrightarrow{F_1} = \overrightarrow{F_2} = 0$

(D) $\overrightarrow{F_1}$ is radially inwards and $\overrightarrow{F_2}$ is radially outards

Q88. JEE Main 2014 (Online) 19th April Morning Slot

Open solution



A square frame of side 10 cm and a long straight wire carrying current 1 A are in the plane of the paper. Starting from close to the wire, the frame moves towards the right with a constant speed of 10 ms^{-1} (see figure). The e.m.f induced at the time the left arm of the frame is at x = 10 cm from the wire is :

(A) 2 μV (B) 1 μV (C) 0.75 μV (D) 0.5 μV

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Q89. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

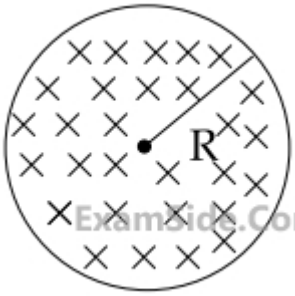
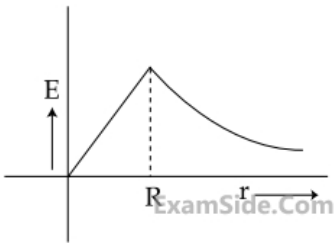
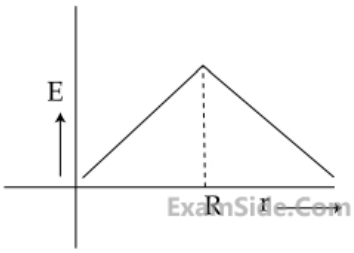


Figure shows a circular area of radius R where a uniform magnetic field $\vec{B}$ is going into the plane of paper and increasing in magnitude at a constant rate. In that case, which of the following graphs, drawn schematically, correctly shows the variation of the induced electric field E(r) ?

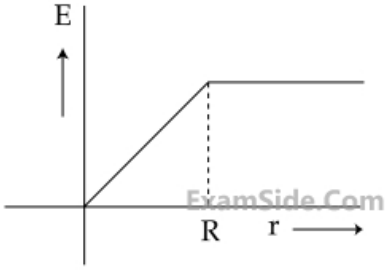
(A)



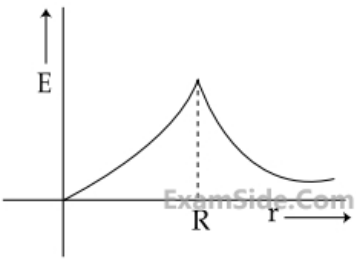
(B)



(C)



(D)



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Q90. JEE Main 2014 (Online) 11th April Morning Slot

A coil of circular cross-section having 1000 turns and 4 cm^2 face area is placed with its axis parallel to a magnetic field which decreases by 10^{-2} Wb m^{-2} in 0.01 s. The e.m.f. induced in the coil is :

(A) 400 mV (B) 200 mV (C) 4 mV (D) 0.4 mV

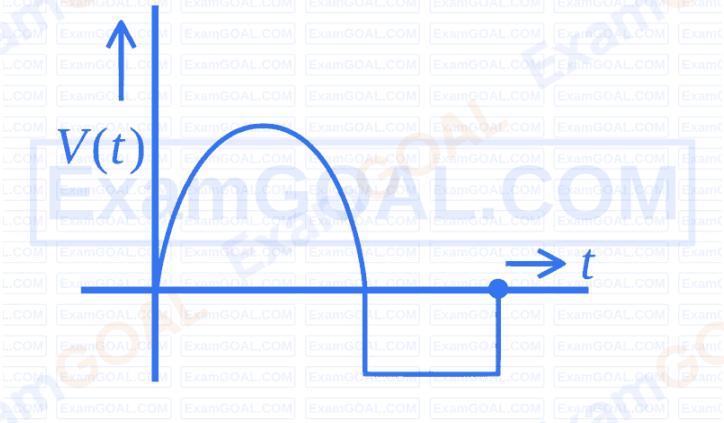
Open solution

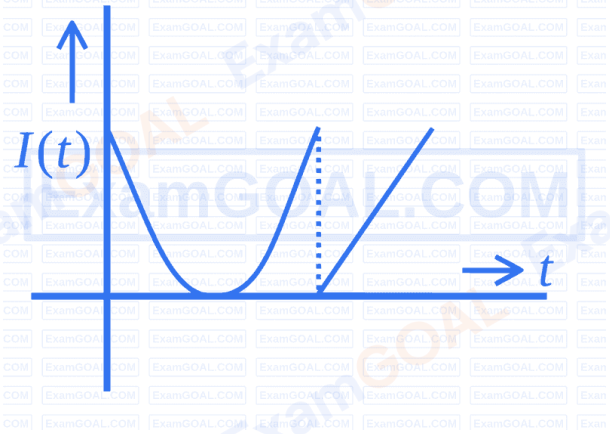
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Q91. JEE Main 2013 (Online) 9th April Morning Slot

Two coils, X and Y, are kept in close vicinity of each other. When a varying current, $I(t)$, flows through coil X, the induced emf $(V(t))$ in coil Y, varies in the manner shown here. The variation of $I(t)$, with time, can then be represented by the graph labelled as graph :

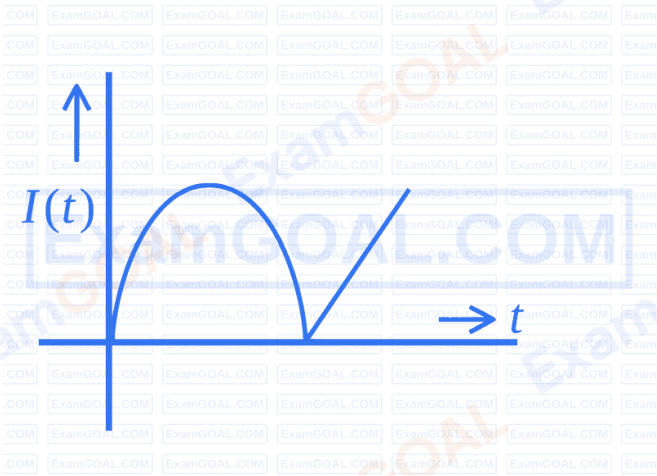
Open solution



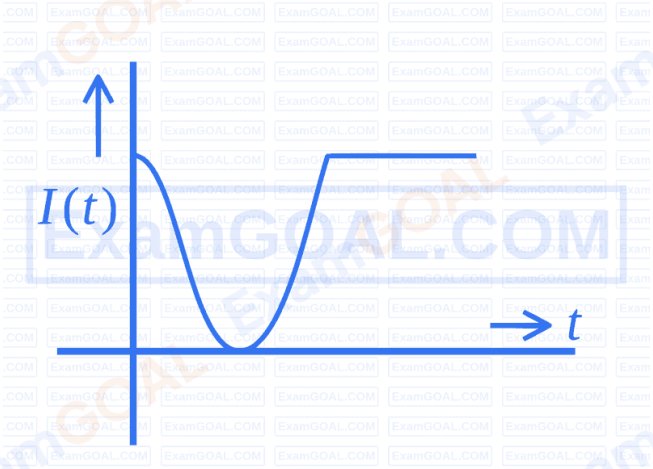


(A)

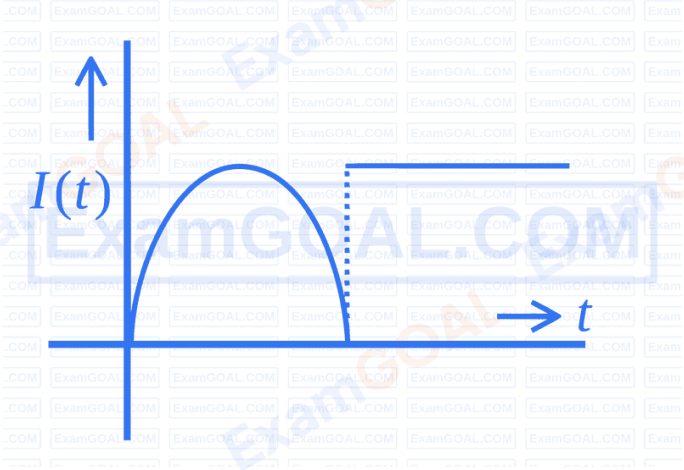
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(B)



(C)



(D)

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(A) A (B) C (C) B (D) D

Q92. JEE Main 2013 (Offline)

A circular loop of radius $0.3\ cm$ lies parallel to a much bigger circular loop of radius $20\ cm$. The center of the small loop is on the axis of the bigger loop. The distance between their centers is $15\ cm$. If a current of $2.0\ A$ flows through the smaller loop, then the flux linked with bigger loop is

Open solution

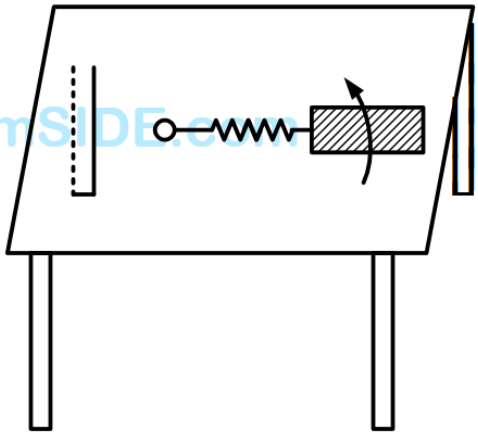
(A) 9.1×10^{-11} weber (B) 6×10^{-11} weber

(C) 3.3×10^{-11} weber (D) 6.6×10^{-9} weber

Q93. JEE Main 2013 (Offline)

A metallic rod of length $'\ell'$ is tied to a string of length 2ℓ and made to rotate with angular speed ω on a horizontal table with one end of the string fixed. If there is a vertical magnetic field $'B'$ in the region, the $e.m.f$ induced across the ends of the rod is

Open solution



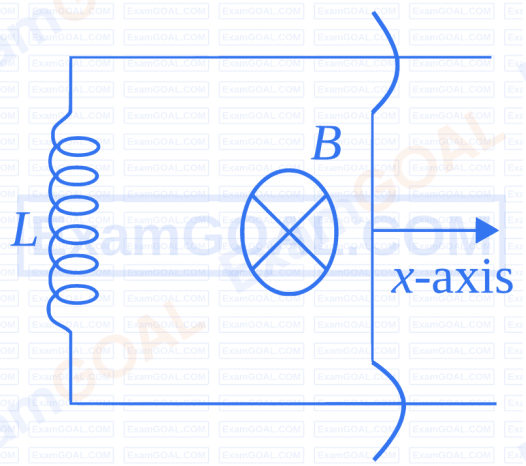
(A) $\frac{2B\omega\ell^2}{2}$ (B) $\frac{3B\omega\ell^2}{2}$ (C) $\frac{4B\omega\ell^2}{2}$ (D) $\frac{5B\omega\ell^2}{2}$

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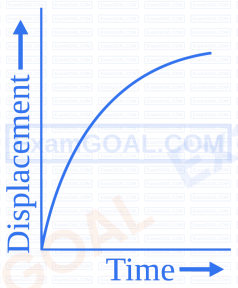
Q94. AIEEE 2012 (Online) 19th May

Open solution

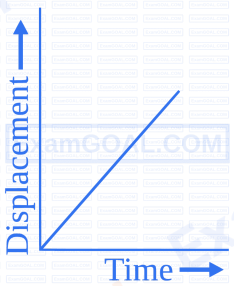
A coil of self inductance L is connected at one end of two rails as shown in figure. A connector of length l, mass m can slide freely over the two parallel rails. The entire set up is placed in a magnetic field of induction B going into the page. At an instant $t = 0$ an initial velocity v_0 is imparted to it and as a result of that it starts moving along x-axis. The displacement of the connector is represented by the figure.



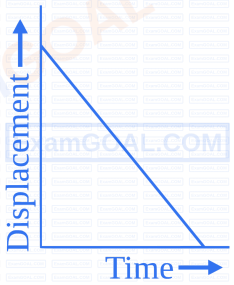
(A)



(B)

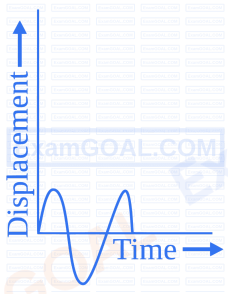


(C)



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(D)



Q95. AIEEE 2012 (Online) 19th May

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: Self inductance of a long solenoid of length L, total number of turns N and radius r is less than $\frac{\pi\mu_0 N^2 r^2}{L}$.
Statement 2: The magnetic induction in the solenoid in Statement 1 carrying current I is $\frac{\mu_0 NI}{L}$ in the middle of the solenoid but becomes less as we move towards its ends.

(A) Statement 1 is true, Statement 2 is false.

(B) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

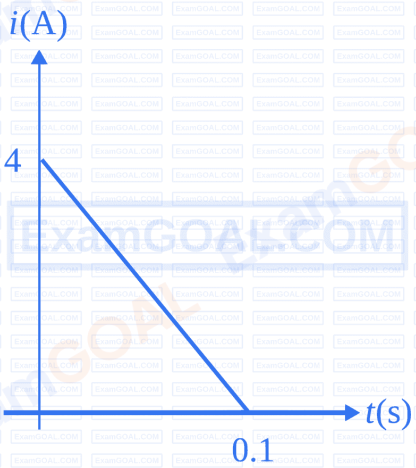
(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

Q96. AIEEE 2012 (Online) 12th May

Open solution

Magnetic flux through a coil of resistance $10\,\Omega$ is changed by $\Delta\phi$ in 0.1 s. The resulting current in the coil varies with time as shown in the figure. Then $|\Delta\phi|$ is equal to (in weber)



(A) 6 (B) 4 (C) 2 (D) 8

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Q97. AIEEE 2012

A coil is suspended in a uniform magnetic field, with the plane of the coil parallel to the magnetic lines of force. When a current is passed through the coil it starts oscillating; It is very difficult to stop. But if an aluminium plate is placed near to the coil, it stops. This is due to :

Open solution

(A) development of air current when the plate is placed

(B) induction of electrical charge on the plate

(C) shielding of magnetic lines of force as aluminium is a para-magnetic material.

(D) electromagnetic induction in the aluminium plate giving rise to electromagnetic damping.

Q98. AIEEE 2011

A boat is moving due east in a region where the earth's magnetic field is $5.0 \times 10^{-5}\ NA^{-1}\,m^{-1}$ due north and horizontal. The boat carries a vertical aerial $2\,m$ long. If the speed of the boat is $1.50\,ms^{-1}$, the magnitude of the induced emf in the wire of aerial is :

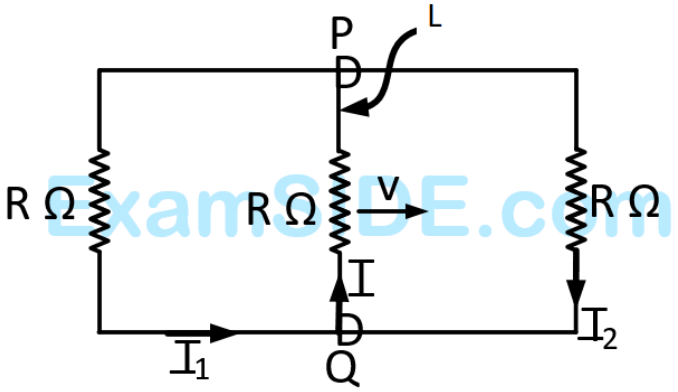
Open solution

(A) $0.75\ mV$ (B) $0.50\ mV$ (C) $0.15\ mV$ (D) $1\ mV$

Q99. AIEEE 2010

A rectangular loop has a sliding connector PQ of length l and resistance $R\,\Omega$ and it is moving with a speed v as shown. The set-up is placed in a uniform magnetic field going into the plane of the paper. The three currents I_1, I_2 and I are

Open solution



(A) $I_1 = -I_2 = \frac{Blv}{6R},\ I = \frac{2Blv}{6R}$ (B) $I_1 = I_2 = \frac{Blv}{3R},\ I = \frac{2Blv}{3R}$

(C) $I_1 = I_2 = I = \frac{Blv}{R}$ (D) $I_1 = I_2 = \frac{Blv}{6R}, I = \frac{Blv}{3R}$

Q100. AIEEE 2008

Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $A = 10\ cm^2$ and length $= 20\,cm$. If one of the solenoid has 300 turns and the other 400 turns, their mutual inductance is
$(\mu_0 = 4\pi \times 10^{-7}\,Tm\,A^{-1})$

Open solution

(A) $2.4\pi \times 10^{-5}H$ (B) $4.8\pi \times 10^{-4}H$ (C) $4.8\pi \times 10^{-5}H$ (D) $2.4\pi \times 10^{-4}H$

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Q101. AIEEE 2007

An ideal coil of $10H$ is connected in series with a resistance of 5Ω and a battery of $5V$. 2 second after the connection is made, the current flowing in ampere in the circuit is

(A) $\left(1-e^{-1}\right)$ (B) $(1-e)$ (C) e (D) e^{-1}

Open solution

Q102. AIEEE 2006

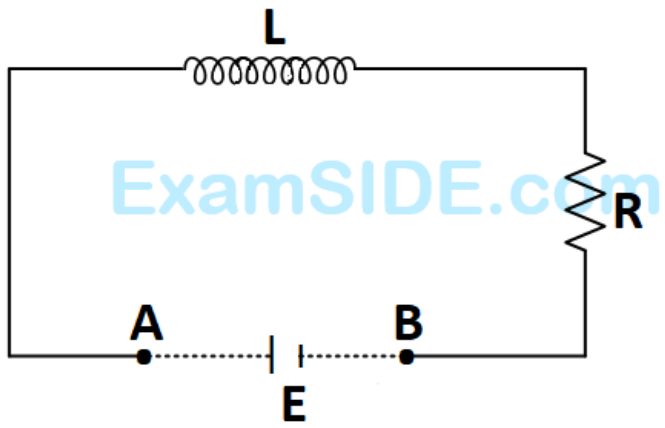
The flux linked with a coil at any instant $'t'$ is given by
$\phi = 10t^2 - 50t + 250$
The induced emf at $t = 3s$ is

(A) $-190\ V$ (B) $-10\ V$ (C) $10\ V$ (D) $190\ V$

Open solution

Q103. AIEEE 2006

An inductor $(L = 100\ mH)$, a resistor $(R = 100\,\Omega)$ and a battery $(E = 100V)$ are initially connected in series as shown in the figure. After a long time the battery is disconnected after short circuiting the points A and B. The current in the circuit $1\ ms$ after the short circuit is



(A) $1/eA$ (B) eA (C) $0.1\ A$ (D) $1\ A$

Open solution

Q104. AIEEE 2005

A coil of inductance $300\ mH$ and resistance $2\,\Omega$ is connected to a source of voltage $2\,V$. The current reaches half of its steady state value in

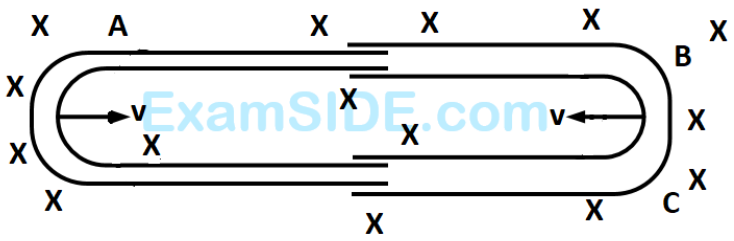
(A) $0.1\ s$ (B) $0.05\ s$ (C) $0.3\ s$ (D) $0.15\ s$

Open solution

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Q105. AIEEE 2005

One conducting U tube can slide inside another as shown in figure, maintaining electrical contacts between the tubes. The magnetic field B is perpendicular to the plane of the figure. If each tube moves towards the other at a constant speed v, then the emf induced in the circuit in terms of B, l and v where l is the width of each tube, will be



(A) $-Blv$ (B) Blv (C) $2\ Blv$ (D) zero

Open solution

Q106. AIEEE 2004

A coil having n turns and resistance $R\Omega$ is connected with a galvanometer of resistance $4R\Omega$. This combination is moved in time t seconds from a magnetic field W_1 weber to W_2 weber. The induced current in the circuit is

(A) $\frac{(W_2-W_1)}{Rnt}$ (B) $-\frac{n(W_2-W_1)}{5\ Rt}$

(C) $-\frac{(W_2-W_1)}{5\ Rnt}$ (D) $-\frac{n(W_2-W_1)}{Rt}$

Open solution

Q107. AIEEE 2004

In a uniform magnetic field of induction B a wire in the form of a semicircle of radius r rotates about the diameter of the circle with an angular frequency ω. The axis of rotation is perpendicular to the field. If the total resistance of the circuit is R, the mean power generated per period of rotation is

(A) $\frac{(B\pi r\omega)^2}{2R}$ (B) $\frac{(B\pi r^2\omega)^2}{8R}$

(C) $\frac{B\pi r^2\omega}{2R}$ (D) $\frac{(B\pi r\omega^2)^2}{8R}$

Open solution

Q108. AIEEE 2004

A metal conductor of length $1\ m$ rotates vertically about one of its ends at angular velocity 5 radians per second. If the horizontal component of earth's magnetic field is $0.2 \times 10^{-4}T$, then the $e.m.f.$ developed between the two ends of the conductor is

(A) $5mV$ (B) $50\mu V$ (C) $5\mu V$ (D) $50mV$

Open solution

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Q109. AIEEE 2003

Two coils are placed close to each other. The mutual inductance of the pair of coils depends upon

(A) the rates at which currents are changing in the two coils

(B) relative position and orientation of the two coils

(C) the currents in the two coils

(D) the materials of the wires of the coils

Open solution

Q110. AIEEE 2003

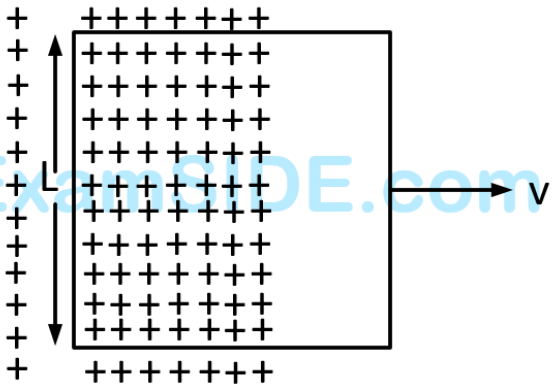
When the current changes from $+2A$ to $-2A$ in 0.05 second, an $e.m.f.$ of $8\ V$ is inducted in a coil. The coefficient of self- induction of the coil is

(A) $0.2H$ (B) $0.4H$ (C) $0.8\ H$ (D) $0.1\ H$

Open solution

Q111. AIEEE 2002

A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B constant in time and space, pointing perpendicular and into the plane at the loop exists everywhere with half the loop outside the field, as shown in figure. The induced emf is



(A) zero (B) RvB (C) vBL/R (D) vBL

Open solution

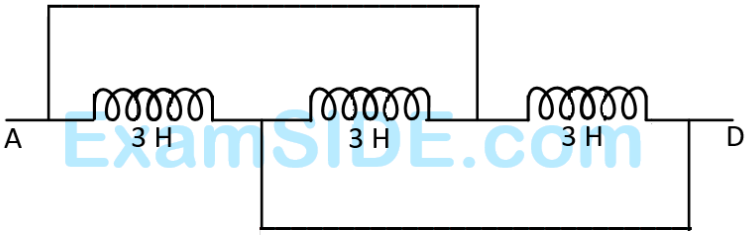
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Q112. AIEEE 2002

The inductance between A and D is

Open solution



(A) $3.66\ H$ (B) $9\ H$ (C) $0.66\ H$ (D) $1\ H$

Numerical 46 questions

Q113. JEE Main 2026 (Online) 5th April Evening Shift

A circular loop of radius 20 cm and resistance 2Ω is placed in a time varying magnetic field $\vec{B} = (2t^2 + 2t + 3)T$. At $t = 0$, for the plane of the loop being perpendicular to the magnetic field and, the induced current in the loop at $t = 3$ s is $\frac{\alpha}{50}$ A. The value of α is ____. (Take $\pi = 22/7$)

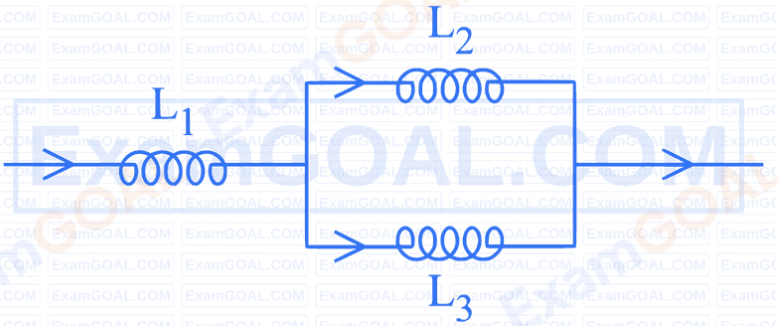
Open solution

Q114. JEE Main 2026 (Online) 5th April Morning Shift

In the given circuit below inductance values of L_1, L_2 and L_3 are same. The magnetic energy stored in the entire circuit is (U_t) and that stored in the L_2 inductor is (U_l). U_t/U_l is ____.

(Ignore the mutual inductance if any)

Open solution



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Q115. JEE Main 2026 (Online) 23rd January Morning Shift

A simple pendulum made of mass 10 g and a metallic wire of length 10 cm is suspended vertically in a uniform magnetic field of 2 T . The magnetic field direction is perpendicular to the plane of oscillations of the pendulum. If the pendulum is released from an angle of 60° with vertical, then maximum induced EMF between the point of suspension and point of oscillation is

____ mV . (Take $g = 10\ m/s^2$)

Open solution

Q116. JEE Main 2026 (Online) 22nd January Evening Shift

A conducting circular loop is rotated about its diameter at a constant angular speed of $100 rad/s$ in a magnetic field of 0.5 T perpendicular to the axis of rotation. When the loop is rotated by 30° from the horizontal position, the induced EMF is 15.4 mV . The radius of the loop is ____ mm.

$\left(\text{Take } \pi = \frac{22}{7}\right)$

Open solution

Q117. JEE Main 2026 (Online) 22nd January Morning Shift

Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω. The energy density in the inductor when the current reaches $\left(\frac{1}{e}\right)$ of its maximum value is $\alpha\pi \times \frac{1}{e^2}\ J/m^3$. The value of α is

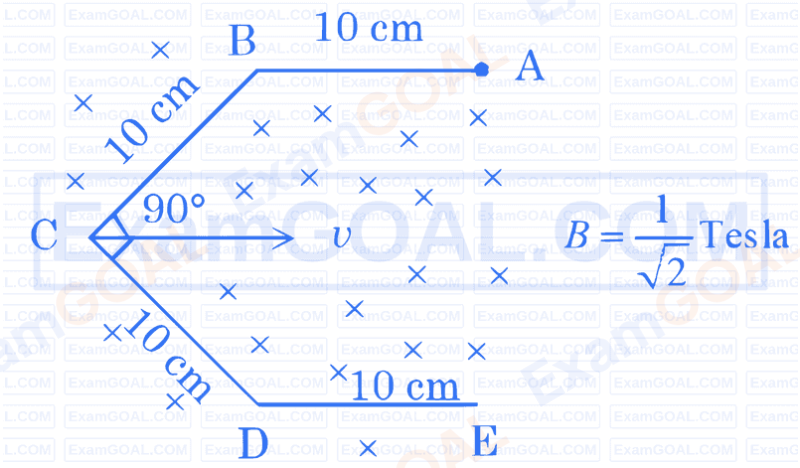
____ .

$(\mu_0 = 4\pi \times 10^{-7} Tm/A)$.

Open solution

Q118. JEE Main 2025 (Online) 4th April Morning Shift

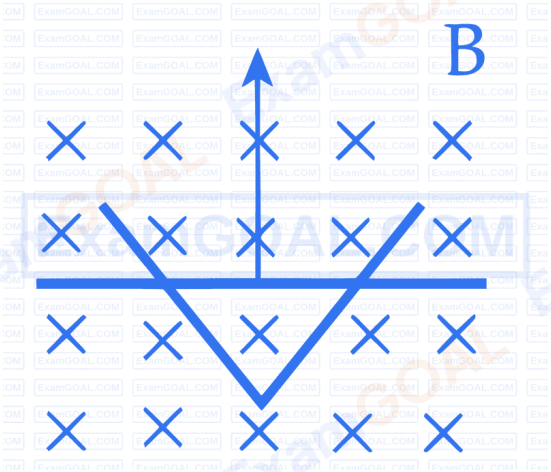
Conductor wire ABCDE with each arm 10 cm in length is placed in magnetic field of $\frac{1}{\sqrt{2}}$ Tesla, perpendicular to its plane. When conductor is pulled towards right with constant velocity of $10\ cm/s$, induced emf between points A and E is _____ mV .



Open solution

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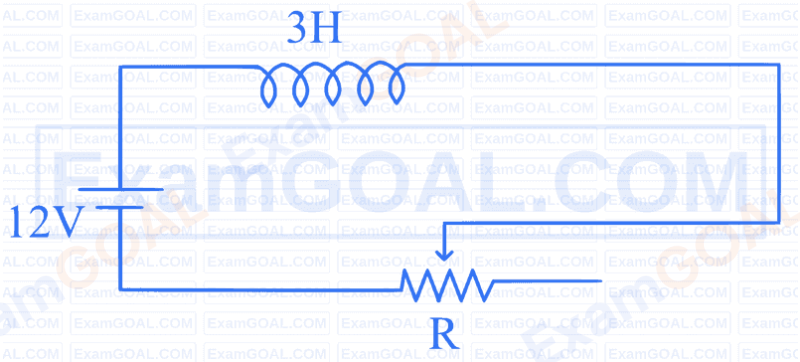
Q119. JEE Main 2025 (Online) 28th January Evening Shift



Open solution

A conducting bar moves on two conducting rails as shown in the figure. A constant magnetic field B exists into the page. The bar starts to move from the vertex at time t = 0 with a constant velocity. If the induced EMF is $E \propto t^n$, then value of n is _____.

Q120. JEE Main 2025 (Online) 23rd January Morning Shift



Open solution

In the given circuit the sliding contact is pulled outwards such that electric current in the circuit changes at the rate of $8\ \mathrm{A/s}$. At an instant when R is 12Ω, the value of the current in the circuit will be _____ A.

Q121. JEE Main 2024 (Online) 5th April Evening Shift

The current in an inductor is given by $\mathrm{I} = (3\mathrm{t} + 8)$ where t is in second. The magnitude of induced emf produced in the inductor is $12\ \mathrm{mV}$. The self-inductance of the inductor _____ mH.

Open solution

Q122. JEE Main 2024 (Online) 4th April Evening Shift

A rod of length $60\ \mathrm{cm}$ rotates with a uniform angular velocity $20\ \mathrm{rads}^{-1}$ about its perpendicular bisector, in a uniform magnetic field $0.5T$. The direction of magnetic field is parallel to the axis of rotation. The potential difference between the two ends of the rod is _____ V.

Open solution

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Q123. JEE Main 2024 (Online) 1st February Evening Shift

A coil of 200 turns and area $0.20\ \mathrm{m}^2$ is rotated at half a revolution per second and is placed in uniform magnetic field of $0.01\ \mathrm{T}$ perpendicular to axis of rotation of the coil. The maximum voltage generated in the coil is $\frac{2\pi}{\beta}$ volt. The value of β is _____.

Open solution

Q124. JEE Main 2024 (Online) 1st February Morning Shift

A rectangular loop of sides $12\ \mathrm{cm}$ and $5\ \mathrm{cm}$, with its sides parallel to the x-axis and y-axis respectively, moves with a velocity of $5\ \mathrm{cm/s}$ in the positive x axis direction, in a space containing a variable magnetic field in the positive z direction. The field has a gradient of $10^{-3}\ \mathrm{T/cm}$ along the negative x direction and it is decreasing with time at the rate of $10^{-3}\ \mathrm{T/s}$. If the resistance of the loop is $6\ \mathrm{m\Omega}$, the power dissipated by the loop as heat is _____ $\times 10^{-9}\ \mathrm{W}$.

Open solution

Q125. JEE Main 2024 (Online) 31st January Evening Shift

The magnetic flux ϕ (in weber) linked with a closed circuit of resistance 8Ω varies with time (in seconds) as $\phi = 5t^2 - 36t + 1$. The induced current in the circuit at $t = 2\ \mathrm{s}$ is _____ A.

Open solution

Q126. JEE Main 2024 (Online) 31st January Morning Shift

A small square loop of wire of side l is placed inside a large square loop of wire of side $L\ (L = l^2)$. The loops are coplanar and their centers coincide. The value of the mutual inductance of the system is $\sqrt{x} \times 10^{-7}\mathrm{H}$, where $x =$ _____.

Open solution

Q127. JEE Main 2024 (Online) 30th January Morning Shift

A ceiling fan having 3 blades of length $80\ \mathrm{cm}$ each is rotating with an angular velocity of 1200 rpm. The magnetic field of earth in that region is $0.5\mathrm{G}$ and angle of dip is 30°. The emf induced across the blades is $\mathrm{N}\pi \times 10^{-5}\ \mathrm{V}$. The value of N is _____.

Open solution

Q128. JEE Main 2024 (Online) 29th January Evening Shift

A horizontal straight wire $5\ \mathrm{m}$ long extending from east to west is falling freely at right angle to horizontal component of earths magnetic field $0.60 \times 10^{-4}\ \mathrm{Wbm}^{-2}$. The instantaneous value of emf induced in the wire when its velocity is $10\ \mathrm{ms}^{-1}$ is _____ $\times 10^{-3}\ \mathrm{V}$.

Open solution

Q129. JEE Main 2024 (Online) 29th January Morning Shift

A square loop of side $10\ \mathrm{cm}$ and resistance 0.7Ω is placed vertically in east-west plane. A uniform magnetic field of $0.20T$ is set up across the plane in north east direction. The magnetic field is decreased to zero in $1\ \mathrm{s}$ at a steady rate. Then, magnitude of induced emf is $\sqrt{x} \times 10^{-3}\ \mathrm{V}$. The value of x is _____.

Open solution

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Q130. JEE Main 2024 (Online) 27th January Morning Shift

Two coils have mutual inductance $0.002\ \mathrm{H}$. The current changes in the first coil according to the relation $\mathrm{i} = \mathrm{i}_0 \sin \omega \mathrm{t}$, where $\mathrm{i}_0 = 5\ \mathrm{A}$ and $\omega = 50\pi$ rad/s. The maximum value of emf in the second coil is $\frac{\pi}{\alpha}\ \mathrm{V}$. The value of α is _____.

Open solution

Q131. JEE Main 2023 (Online) 15th April Morning Shift

A $20\ \mathrm{cm}$ long metallic rod is rotated with $210\ \mathrm{rpm}$ about an axis normal to the rod passing through its one end. The other end of the rod is in contact with a circular metallic ring. A constant and uniform magnetic field $0.2\ \mathrm{T}$ parallel to the axis exists everywhere. The emf developed between the centre and the ring is _____ mV.

Take $\pi = \frac{22}{7}$

Open solution

Q132. JEE Main 2023 (Online) 13th April Evening Shift

An insulated copper wire of 100 turns is wrapped around a wooden cylindrical core of the cross-sectional area $24\ \mathrm{cm}^2$. The two ends of the wire are connected to a resistor. The total resistance in the circuit is $12\ \Omega$. If an externally applied uniform magnetic field in the core along its axis changes from $1.5\ \mathrm{T}$ in one direction to $1.5\ \mathrm{T}$ in the opposite direction, the charge flowing through a point in the circuit during the change of magnetic field will be _____ mC.

Open solution

Q133. JEE Main 2023 (Online) 12th April Morning Shift

A conducting circular loop is placed in a uniform magnetic field of $0.4\ \mathrm{T}$ with its plane perpendicular to the field. Somehow, the radius of the loop starts expanding at a constant rate of $1\ \mathrm{mm/s}$. The magnitude of induced emf in the loop at an instant when the radius of the loop is $2\ \mathrm{cm}$ will be _____ $\mu\mathrm{V}$.

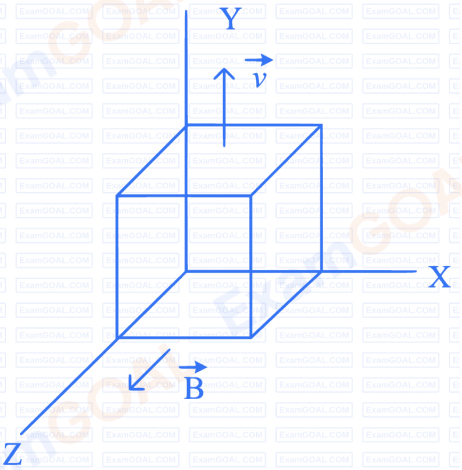
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Q134. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

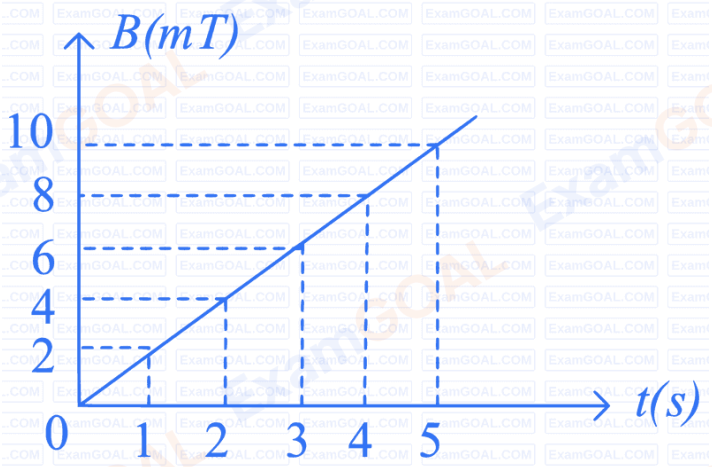
A metallic cube of side $15\ \text{cm}$ moving along y-axis at a uniform velocity of $2\ \text{ms}^{-1}$. In a region of uniform magnetic field of magnitude $0.5\ \text{T}$ directed along z-axis. In equilibrium the potential difference between the faces of higher and lower potential developed because of the motion through the field will be _____ mV.



Q135. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

The magnetic field B crossing normally a square metallic plate of area $4\ \text{m}^2$ is changing with time as shown in figure. The magnitude of induced emf in the plate during $\text{t} = 2s$ to $\text{t} = 4s$, is _____ mV.



Q136. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

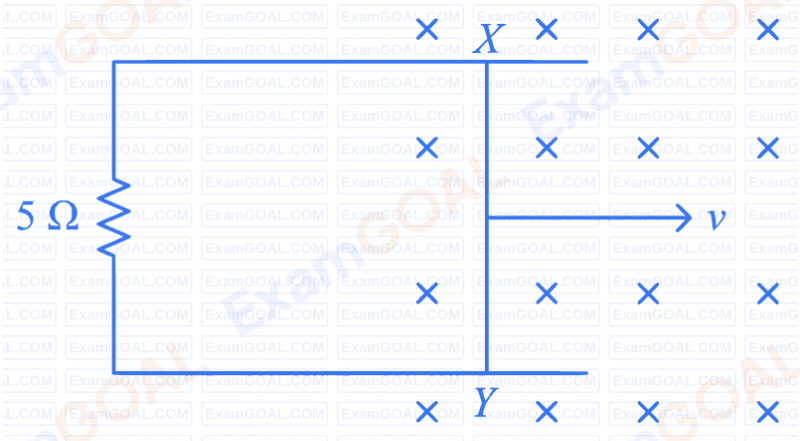
A square loop of side $2.0\ \text{cm}$ is placed inside a long solenoid that has 50 turns per centimetre and carries a sinusoidally varying current of amplitude $2.5\ \text{A}$ and angular frequency $700\ \text{rad s}^{-1}$. The central axes of the loop and solenoid coincide. The amplitude of the emf induced in the loop is $x \times 10^{-4}\ \text{V}$. The value of x is _____.

(Take, $\pi = \frac{22}{7}$)

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Q137. JEE Main 2023 (Online) 10th April Morning Shift

A 1 m long metal rod XY completes the circuit as shown in figure. The plane of the circuit is perpendicular to the magnetic field of flux density 0.15 T. If the resistance of the circuit is 5Ω, the force needed to move the rod in direction, as indicated, with a constant speed of 4 m/s will be _____ 10^{-3} N.



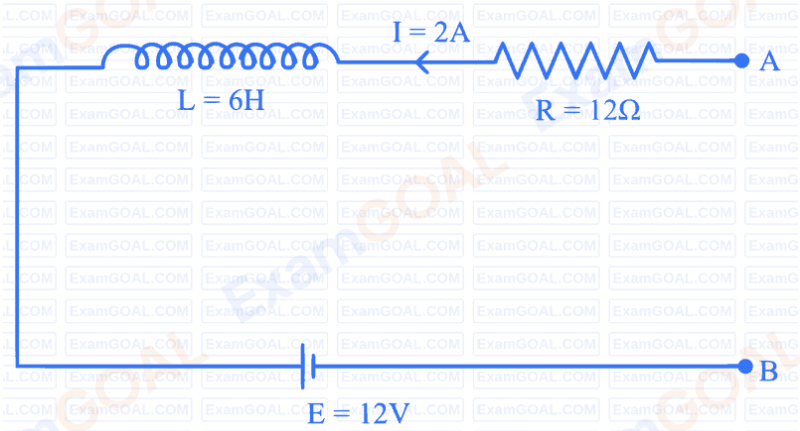
Open solution

Q138. JEE Main 2023 (Online) 6th April Evening Shift

Two concentric circular coils with radii $1\ \text{cm}$ and $1000\ \text{cm}$, and number of turns 10 and 200 respectively are placed coaxially with centers coinciding. The mutual inductance of this arrangement will be _____ $\times 10^{-8}\text{H}$. (Take, $\pi^2 = 10$)

Open solution

Q139. JEE Main 2023 (Online) 30th January Morning Shift



As per the given figure, if $\frac{\text{dI}}{\text{dt}} = -1\ \text{A}/s$ then the value of V_{AB} at this instant will be _____ V.

Open solution

Q140. JEE Main 2023 (Online) 29th January Morning Shift

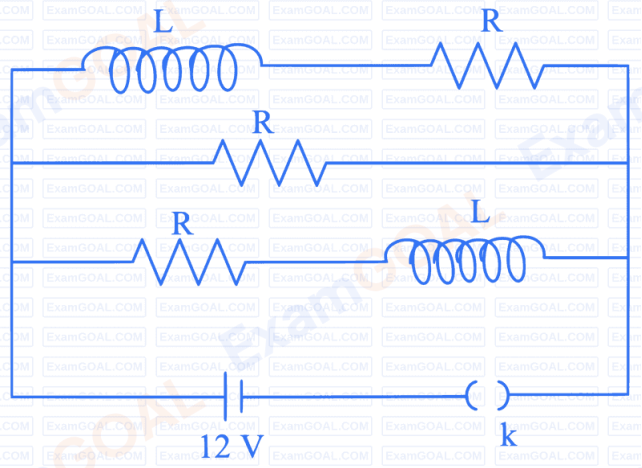
A certain elastic conducting material is stretched into a circular loop. It is placed with its plane perpendicular to a uniform magnetic field B = 0.8 T. When released the radius of the loop starts shrinking at a constant rate of 2 cms^{-1}. The induced emf in the loop at an instant when the radius of the loop is 10 cm will be _____ mV.

Open solution

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Q141. JEE Main 2023 (Online) 24th January Evening Shift

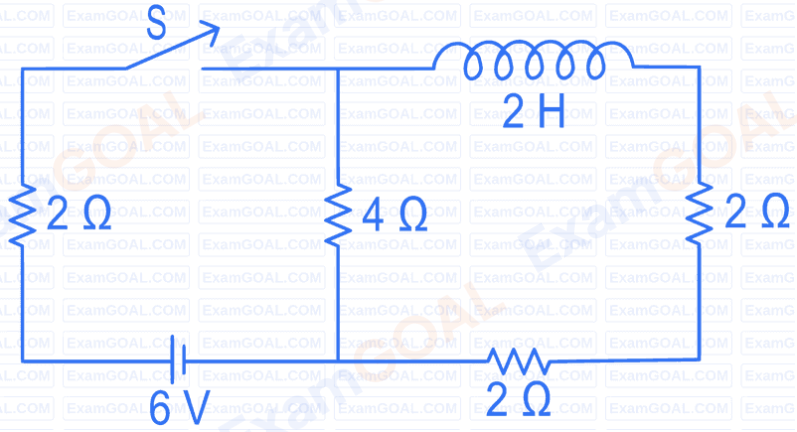
Three identical resistors with resistance R = 12Ω and two identical inductors with self inductance L = 5 mH are connected to an ideal battery with emf of 12 V as shown in figure. The current through the battery long after the switch has been closed will be _____ A.



Open solution

Q142. JEE Main 2022 (Online) 28th July Evening Shift

For the given circuit the current through battery of 6 V just after closing the switch 'S' will be _____ A.



Open solution

Q143. JEE Main 2022 (Online) 27th July Evening Shift

A conducting circular loop is placed in $X-Y$ plane in presence of magnetic field $\vec{B}=\left(3t^3\,\hat{j}+3t^2\,\hat{k}\right)$ in SI unit. If the radius of the loop is $1\ \mathrm{m}$, the induced emf in the loop, at time, $t=2\ \mathrm{s}$ is $n\pi\ \mathrm{V}$. The value of n is _____.

Open solution

Q144. JEE Main 2022 (Online) 26th July Evening Shift

In a coil of resistance $8\,\Omega$, the magnetic flux due to an external magnetic field varies with time as $\phi=\frac{2}{3}\left(9-t^2\right)$. The value of total heat produced in the coil, till the flux becomes zero, will be _____ J.

Open solution

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Q145. JEE Main 2022 (Online) 25th July Evening Shift

Magnetic flux (in weber) in a closed circuit of resistance 20 Ω varies with time t(s) at $\phi = 8t^2 - 9t + 5$. The magnitude of the induced current at t = 0.25 s will be _____ mA.

Open solution

Q146. JEE Main 2022 (Online) 27th June Evening Shift

A metallic rod of length 20 cm is placed in North-South direction and is moved at a constant speed of 20 m/s towards East. The horizontal component of the Earth's magnetic field at that place is 4×10^{-3} T and the angle of dip is 45°. The emf induced in the rod is _____ mV.

Open solution

Q147. JEE Main 2022 (Online) 26th June Morning Shift

A 10 Ω, 20 mH coil carrying constant current is connected to a battery of 20 V through a switch. Now after switch is opened current becomes zero in 100 μs. The average e.m.f. induced in the coil is _____ V.

Open solution

Q148. JEE Main 2022 (Online) 25th June Morning Shift

The current in a coil of self inductance 2.0 H is increasing according to $I = 2 \sin(t^2)$ A. The amount of energy spent during the period when current changes from 0 to 2 A is _____ J.

Open solution

Q149. JEE Main 2022 (Online) 24th June Evening Shift

A circular coil of 1000 turns each with area $1m^2$ is rotated about its vertical diameter at the rate of one revolution per second in a uniform horizontal magnetic field of 0.07T. The maximum voltage generation will be _____ V.

Open solution

Q150. JEE Main 2021 (Online) 26th August Evening Shift

A circular coil of radius 8.0 cm and 20 turns is rotated about its vertical diameter with an angular speed of 50 rad s^{-1} in a uniform horizontal magnetic field of 3.0×10^{-2} T. The maximum emf induced the coil will be ..... $\times 10^{-2}$ volt (rounded off to the nearest integer)

Open solution

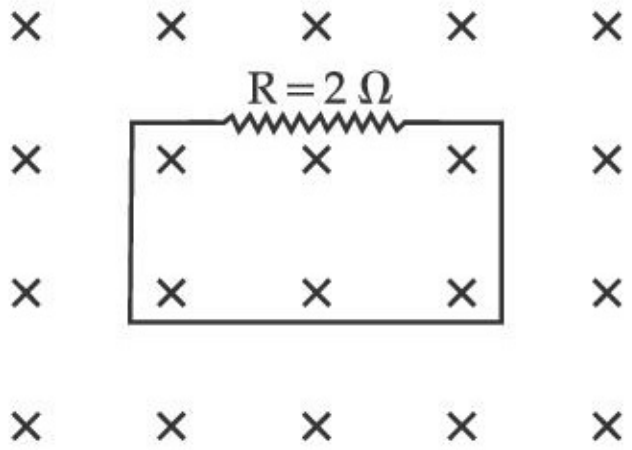
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Q151. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

In the given figure the magnetic flux through the loop increases according to the relation $\phi_B(t) = 10t^2 + 20t$, where ϕ_B is in milliwebers and t is in seconds.

The magnitude of current through R = 2Ω resistor at t = 5 s is _____ mA.



Q152. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

A circular conducting coil of radius 1 m is being heated by the change of magnetic field $\vec{B}$ passing perpendicular to the plane in which the coil is laid. The resistance of the coil is 2 $\mu\Omega$. The magnetic field is slowly switched off such that its magnitude changes in time as

$$B = \frac{4}{\pi} \times 10^{-3} T \left(1 - \frac{t}{100}\right)$$

The energy dissipated by the coil before the magnetic field is switched off completely is E = _____ mJ.

Q153. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

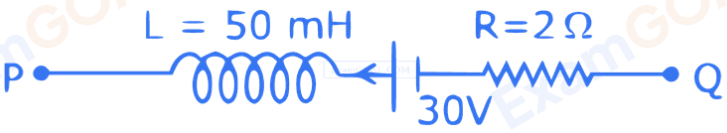
A coil of inductance 2 H having negligible resistance is connected to a source of supply whose voltage is given by V = 3t volt. (where t is in second). If the voltage is applied when t = 0, then the energy stored in the coil after 4 s is _____J.

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Q154. JEE Main 2020 (Online) 6th September Morning Slot

A part of a complete circuit is shown in the figure. At some instant, the value of current I is 1A and it is decreasing at a rate of 10^2 A s^{-1}. The value of the potential difference $V_P - V_Q$, (in volts) at that instant, is _____.

Open solution



Q155. JEE Main 2020 (Online) 5th September Morning Slot

Two concentric circular coils, C_1 and C_2 are placed in the XY plane. C_1 has 500 turns, and a radius of 1 cm. C_2 has 200 turns and radius of 20 cm. C_2 carries a time dependent current $I(t) = (5t^2 - 2t + 3)$ A where t is in s. The emf induced in C_1 (in mV), at the instant t = 1 s is $\frac{4}{x}$. The value of x is ___ .

Open solution

Q156. JEE Main 2020 (Online) 2nd September Morning Slot

A circular coil of radius 10 cm is placed in a uniform magnetic field of 3.0×10^{-5} T with its plane perpendicular to the field initially. It is rotated at constant angular speed about an axis along the diameter of coil and perpendicular to magnetic field so that it undergoes half of rotation in 0.2 s. The maximum value of EMF induced (in μV) in the coil will be close to the integer _____.

Open solution

Q157. JEE Main 2020 (Online) 9th January Morning Slot

In a fluorescent lamp choke (a small transformer) 100 V of reverse voltage is produced when the choke current changes uniformly from 0.25 A to 0 in a duration of 0.025 ms. The self-inductance of the choke (in mH) is estimated to be _____.

Open solution

Q158. JEE Main 2020 (Online) 7th January Morning Slot

A loop ABCDEFA of straight edges has six corner points A(0, 0, 0), B(5, 0, 0), C(5, 5, 0), D (0, 5, 0), E(0, 5, 5) and F(0, 0, 5). The magnetic field in this region is $\vec{B} = \left(3\hat{i} + 4\hat{k}\right)T$. The quantity of flux through the loop ABCDEFA (in Wb) is _____.

Open solution

Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q159. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

Open solution

A fighter plane of length 20 m, wing span (distance from tip of one wing to the tip of the other wing) of 15 m and height 5 m is flying towards east over Delhi. Its speed is 240 ms^{-1}. The earth's magnetic field over Delhi is 5×10^{-5} T with the declination angle ~ 0° and dip of θ such that $\sin\theta = 2/3$. If the voltage developed is V_B between the lower and upper side of the plane and V_W between the tips of the wings then V_B and V_W are close to :

(A) V_B = 45 mV; V_W = 120 mV with right side of pilot at higher voltage

(B) V_B = 45 mV; V_W = 120 mV with left side of pilot at higher voltage

(C) V_B = 40 mV; V_W = 135 mV with right side of pilot at high voltage

(D) V_B = 40 mV; V_W = 135 mV with left side of pilot at higher voltage

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **D**	3. **A**	4. **A**	5. **D**
6. **A**	7. **C**	8. **D**	9. **C**	10. **C**
11. **A**	12. **B**	13. **C**	14. **B**	15. **A**
16. **D**	17. **D**	18. **C**	19. **C**	20. **B**
21. **A**	22. **D**	23. **B**	24. **B**	25. **C**
26. **A**	27. **C**	28. **C**	29. **D**	30. **A**
31. **A**	32. **B**	33. **A**	34. **B**	35. **B**
36. **C**	37. **A**	38. **C**	39. **C**	40. **B**
41. **A**	42. **C**	43. **D**	44. **D**	45. **B**
46. **A**	47. **B**	48. **A**	49. **A**	50. **C**
51. **C**	52. **D**	53. **B**	54. **C**	55. **D**
56. **B**	57. **D**	58. **D**	59. **C**	60. **C**
61. **D**	62. **C**	63. **A**	64. **C**	65. **D**
66. **D**	67. **D**	68. **B**	69. **C**	70. **A**
71. **B**	72. **C**	73. **D**	74. **D**	75. **B**
76. **D**	77. **B**	78. **A**	79. **C**	80. **C**
81. **A**	82. **A**	83. **D**	84. **B**	85. **A**
86. **B**	87. **C**	88. **B**	89. **A**	90. **A**
91. **B**	92. **A**	93. **D**	94. **D**	95. **B**
96. **C**	97. **D**	98. **C**	99. **B**	100. **D**
101. **A**	102. **B**	103. **A**	104. **A**	105. **C**
106. **B**	107. **B**	108. **B**	109. **B**	110. **D**
111. **D**	112. **D**			

Numerical

113. **44**	114. **6**	115. **100**	116. **14**	117. **20**
118. **10**	119. **1**	120. **3**	121. **4**	122. **0**
123. **5**	124. **216**	125. **2**	126. **128**	127. **32**
128. **3**	129. **2**	130. **2**	131. **88**	132. **60**
133. **50**	134. **150**	135. **8**	136. **44**	137. **18**
138. **4**	139. **30**	140. **10**	141. **3**	142. **1**

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143. **12**	144. **2**	145. **250**	146. **16**	147. **400**
148. **4**	149. **440**	150. **60**	151. **60**	152. **80**
153. **144**	154. **33**	155. **5**	156. **15**	157. **10**
158. **175**				

Out of syllabus

159. **B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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